
Minimax Estimation–Regret Tradeoffs and Certified Exploration in Performative Prediction

Shriraghav Ashok*
Haas School of Business
University of California, Berkeley
Berkeley, California 94720
sashok24@berkeley.edu

Vignesh Nagarajan*
Brown Scholars Fund
Texas A&M University
College Station, Texas 77843
vigneshn26@tamu.edu

Abstract

Deployed machine learning models can change the environments that generate their future training data. Learning how an environment responds therefore requires exploration, but these exploratory deployments can reduce performance. Existing approaches to performative prediction typically estimate distributional responses without explicitly accounting for the cost of generating the perturbations needed to learn them. This work studies this tradeoff in a structural market setting where a dealer’s quoted spread affects the adverse order flow it subsequently observes. An exact conservation law links uncertainty in the estimated response slope to cumulative performative regret under any stationary exploration policy. Building on this result, minimax lower bounds and the optimal exploration rate are characterized, and an adaptive procedure updates only when an anytime confidence sequence certifies retraining stability. The theory is validated through 36 preregistered pipeline checks and a public corporate-bond calibration that matches published results across all 10 evaluation cells within 0.8%. These results identify a fundamental limit of performative learning: when deployment changes the data distribution, learning that response necessarily comes with a performance cost.

1 Introduction

When a deployed model changes the distribution it is trained on, retraining becomes a fixed-point iteration: performative prediction [Perdomo et al., 2020, Hardt and Mendler-Dünner, 2023] shows repeated risk minimization converges when the distribution’s sensitivity ε is small against the loss curvature γ ($m = \varepsilon\beta/\gamma < 1$), and converges to a *performatively stable* point that is not the *performative optimum*. Algorithms that close this gap, performative gradient descent and its descendants [Izzo et al., 2021, Miller et al., 2021, Mendler-Dünner et al., 2020, Drusvyatskiy and Xiao, 2023, Li and Wai, 2022, Brown et al., 2022], all consume an estimate of the distribution’s response $dD/d\theta$, obtained by perturbing deployments. The literature treats that estimate’s accuracy as an assumption and the perturbations as free. Neither is: the perturbations are deployments of a real decision variable, and their cost lands in the decision-maker’s own objective. *What does it cost to learn your own influence?* This paper answers with an identity, a floor, a design rule, and an agent.

Contributions.

1. **A conservation law (§3).** Along any stationary exploration policy, $\text{Var}(\hat{\varepsilon}) \times C_T = \frac{1}{2}\gamma_{\text{PO}}\sigma^2$: slope uncertainty times cumulative performative regret is a policy-independent invariant (Theorem 1). The anchoring is delicate: anchored at the performative optimum instead, the product is *false*, inflated by a verified factor.

*Equal contribution.

2. **Minimax floors, and their exact reach (§3).** No non-anticipating policy beats the floor: exactly under deviation budgets; within $\leq 13.5\times$ under value budgets, sharp constant numerically supported but open (Theorem 2). The floor is also *structure-proof within an exact reach* (Theorem 3): within $2/c$ of the anchor even full knowledge of the family cannot beat it; one probe beyond, an explicit design does. The trust region keeps the agent inside the protection.
3. **Optimal exploration design (§3).** Under a budget priced by the decision-maker’s own curvature, the optimal exploration covariance is $\Gamma_{\text{PO}}^{-1/2}$ -shaped; isotropic exploration overpays by the dispersion factor $F = d \operatorname{tr} \Gamma / (\operatorname{tr} \Gamma^{1/2})^2$: an inversion of “naive exploration is optimal” for LQR [Simchowitz and Foster, 2020]; a support-degeneracy theorem and a misspecification crossover complete the picture.
4. **A certified agent (§3–4).** SafeD-PerfGD explores on a D-optimal support, fits an anchored response family with an anytime confidence sequence, and corrects only when a perturbed-modulus certificate guarantees closed-loop stability; otherwise it freezes the correction while the provably-stable blind step keeps running. It reaches the performative optimum where blind retraining stalls; no published baseline Pareto-dominates it on median (final error, regret).

Positioning. Jagadeesan et al. [2022] bound the regret of *finding* the performative optimum; we price *knowing your own feedback*: lower bounds and an invariance identity where they give algorithmic upper bounds, with no retraining loop, stability constraint, or design shaping in their model. Bracale et al. [2025] design deployments to learn the distribution map and bound the regret of doing so at an order level; we price the same activity exactly, in the objective’s own curvature, and close it from below. Lin and Zrnic [2024] show misspecified structural models still help performative optimization; our crossover theorem sharpens that message into a budget-explicit rule. In operations research, incomplete learning under myopic pricing is classical [Rothschild, 1974, Harrison et al., 2012, Keskin and Zeevi, 2014, Broder and Rusmevichientong, 2012, den Boer, 2015]; Lai and Robbins [1979]’s adaptive designs are *points on* our frontier, but the frontier, its floor, and the performative loop generating it do not appear: the demand curve is exogenous. From control we inherit dual control [Feldbaum, 1960] and least-costly identification [Bombois et al., 2006]: the plant is a retraining learner, the budget is the optimizer’s objective, the constraint is loop contraction (lower bounds are van Trees arguments [Gill and Levit, 1995]); task-optimal design [Wagenmaker et al., 2021, Dean et al., 2020] and the classical canon [Kiefer and Wolfowitz, 1960, Pukelsheim, 2006] never price the design in the experimenter’s own objective.

2 Background

The environment is the single-book structural market of REFLEX [Nagarajan and Ashok, 2026], the foundation this paper builds on, its gate and participation constants folded into the flow scales. A dealer quotes half-spread h . Benign flow arrives at rate $U(h) = Ae^{-kh}$; informed (toxic) flow at rate $\tau(h) = C_0 + C_1 e^{-ch}$: tighter quotes summon more toxic flow. The quote-summons-toxicity channel is the dealer’s adverse-selection problem [Guéant et al., 2013]; that its own quotes feed the flow it reads is Barzykin et al. [2026]’s price-reading effect. Profit at frozen toxic level T is $J(h; T) = hU(h) + (h - \psi)T - w(h - h_{\text{ref}})^2$ (ψ : adverse severity; w : quoting-cost convexity); the *performative value* is $\Phi(h) = J(h, \tau(h))$. Write $\varepsilon(h) = -\tau'(h)$ for the response slope (the scalar $dD/d\theta$), $\gamma = -\partial_h^2 J$, and $\gamma_{\text{PO}} = -\Phi'' = \gamma + \varepsilon(2 + c\psi - ch)$. Blind retraining (RRM) is the cobweb $h_{t+1} = \text{BR}(\tau(h_t))$, a contraction iff the modulus $m = \varepsilon\beta/\gamma = \varepsilon/\gamma$ ($\beta = 1$ here) is below one [Perdomo et al., 2020]; it converges to the stable point h_{SP} , while the optimum $h_{\text{PO}} = \arg \max \Phi$ sits at a *tighter* spread here, since toxic flow is profitable net of ψ : the echo-chamber gap [Miller et al., 2021]. Deployments return noisy flow $y_t = \tau(h_t) + \sigma\zeta_t$ (A3). The estimand is ε at the operating point; exploration deviations $d_t = h_t - h^*$ are priced by the *incremental* cost $C_T = \sum_{t \leq T} \mathbb{E}[\Phi(h^*) - \Phi(h_t)]$, anchored at the certainty-equivalent operating point h^* (A1: local quadratic scope; A5: stationary exploration scale). Results are stated for this scalar family and, where flagged, its d -dimensional and LQ instances; full statements and proofs, keyed by register identifiers (T1, T2, ...), are in Appendices A–B.

86 3 Methodology

87 **Free data saturates.** Without deliberate exploration the retraining trajectory’s identifying energy
 88 is finite: for deployment error d_0 and modulus m it is capped at $d_0^2/(1 - m^2)$ (T1; matrix version
 89 by a Lyapunov identity). A corollary was discovered by a failed prediction (§4): retraining on *noisy*
 90 observations converts observation noise to deployment noise with gain $1/\gamma$, adding a stationary
 91 excitation floor $(\sigma/\gamma)^2/(1 - m^2)$ per step; both closed forms verify. The stabler the loop, the less it
 92 reveals: *safety implies blindness*, the performative sharpening of incomplete learning under myopic
 93 policies [Keskin and Zeevi, 2014, Harrison et al., 2012].

94 **The exchange rate.** Deliberate exploration buys information at an exact, policy-independent price.

95 **Theorem 1** (Exchange rate; T2). *Under A1, A2-sym, A3 and A6, for the OLS slope estimator on any*
 96 *stationary exploration policy with deviations $\{d_t\}$,*

$$\text{Var}(\hat{\varepsilon}) \times C_T = \frac{1}{2} \gamma_{\text{PO}} \sigma^2 (1 + o(1)),$$

97 *independent of the exploration amplitude, shape, schedule, and modulus.*

98 The mechanism is cancellation: information scales as the centered energy S_T , cost as $\frac{1}{2} \gamma_{\text{PO}} \sum_t d_t^2$;
 99 for mean-centered exploration these coincide. Anchored at h_{PO} instead, the same product is false
 100 (inflated by an explicit factor, falsified numerically before the incremental version was proved): the
 101 invariant prices *exploration*, not *suboptimality*. Lai and Robbins [1979]’s $O(\log n)$ -cost schemes
 102 are points on this frontier; the identity says *every* stationary policy is. A corollary prices the echo
 103 chamber: closing it pays iff the horizon exceeds an explicit break-even (T8, Appendix).

104 **Minimax floors.** The identity is for OLS-on-stationary policies; the floors are for everyone.

105 **Theorem 2** (Minimax floors; T3–T4). (i) *Over all non-anticipating (randomized, adaptive) policies*
 106 *with pathwise budget $\sum_{t \leq T} d_t^2 \leq D$, all estimators, and any prior of width w : $\inf \sup \text{Var}(\hat{\varepsilon}) \geq$*
 107 *$\sigma^2/(D + \sigma^2(\pi/w)^2)$, so the product floor is attained up to the prior term, with equality by symmetric*
 108 *probing. (ii) Under a budget on realized performative cost with certainty-equivalent anchoring:*
 109 *$\sup \text{Var}(\hat{\varepsilon}) \times C_T \geq \gamma_{\text{PO}} \sigma^2/27 (1 - O(T^{-1/2}))$ over adaptive policies, by a Le Cam two-point*
 110 *reduction with the exploited rebate vanishing at rate $T^{-1/2}$ (Lemma L2, Appendix A); the sharp*
 111 *constant $\frac{1}{2}$ is numerically supported and open (OPEN-1).*

112 Can a policy that *knows the family’s functional form* shop for designs that beat the floor? The answer
 113 has an exact boundary:

114 **Theorem 3** (Structure-proofness, exact reach; T9). *Let $R(\mu)$ be the known-family Cramér–Rao*
 115 *product of design μ , normalized by the floor (a bound for unbiased, and asymptotically all regular,*
 116 *estimation). (i) Every estimable μ supported in $[h^* - 2/c, \infty)$ has $R(\mu) \geq 1$, strict off one two-point*
 117 *configuration, and $\inf_{\mu} R(\mu) = 1$: within reach, knowing the family cannot beat the floor. (ii) The*
 118 *reach is exact: a four-point design with one below-reach probe of weight 3×10^{-4} attains $R = 0.8517$*
 119 *(50-digit verified).*

120 The proof exhibits the family element $\phi_0 = 2/c - e^{-c(h-h^*)}(2/c + h - h^*)$, whose pointwise
 121 domination $|\phi_0| \leq |h - h^*|$ holds exactly on the reach; a Rayleigh reduction converts domination into
 122 the floor. Two boundary phenomena calibrate the scope: with c known the floor breaks outright (ratio
 123 0.30 at $t=1.0$, 0.04 at $t=0.05$), and below the reach the violators are asymmetric near-anchor clusters
 124 with one vanishing-weight far probe, invisible to coarse search (our $1.005\times$ verdict, corrected as
 125 pivot eleven). *The floor is an ignorance-of-curvature phenomenon with an exactly known reach, and*
 126 *the trust region $r \leq 2/c$ keeps the agent inside it.*

127 **Shape (T5).** In d dimensions with curvature $\Gamma_{\text{PO}} \succ 0$ pricing the budget $\frac{1}{2} \mathbb{E}[u^\top \Gamma_{\text{PO}} u]$, the A-
 128 optimal exploration second moment is $M^* \propto \Gamma_{\text{PO}}^{-1/2}$: *explore where your objective is flat*. Isotropic
 129 exploration overpays by exactly the dispersion factor $F = d \text{tr} \Gamma_{\text{PO}} / (\text{tr} \Gamma_{\text{PO}}^{1/2})^2 \in [1, d]$: naive
 130 exploration, optimal for LQR [Simchowitz and Foster, 2020], is suboptimal here by F . **Support**
 131 **(T6).** The exponential family is Chebyshev-degenerate on thin designs: fewer than three support
 132 points (or vanishing spread, or collinear probe schedules) cannot identify (C_0, C_1, c) , a cliff we
 133 hit twice in the wild (§4). **Anchor (T7).** With budget B and horizon T , fitting the structural
 134 family beats nonparametric slope estimation iff the misspecification at the operating point is below
 135 $|\tau'''|B/(3\gamma_{\text{PO}}T)$: Lin and Zrnic [2024]’s message sharpened to a budget-priced rule.

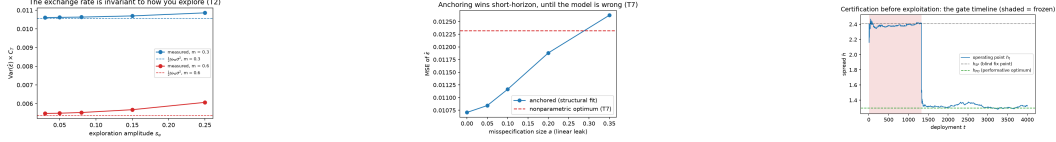


Figure 1: **Left:** $\text{Var}(\hat{\epsilon}) \times C_T$ vs amplitude, two moduli: flat on the invariant (dashed), small drift as A1 predicts. **Middle:** the T7 misspecification crossover. **Right:** certification before exploitation (shaded: frozen): the gate clears after $\sim 1,300$ deployments (freeze fraction 0.33, E4 log) and the agent steps to h_{PO} .

The agent. SafeD-PerfGD composes the three results per deployment round: (i) *explore* on the D-optimal three-point support re-centered at the operating point, trust-region clipped; (ii) *fit* (C_0, C_1, c) by profile least squares with an anytime self-normalized confidence width ci_t for $\hat{\epsilon}$; (iii) *gate*: apply the performative correction $-(h - \psi)\hat{\epsilon}$ with Newton scaling $\eta = 1/\hat{\gamma}_{\text{PO}}$ only if $\eta ci_t L_{\text{fam}} \leq \text{margin}$, where the perturbed-modulus lemma (L4), with the family’s own Lipschitz constant $L_{\text{fam}} = 1 + \hat{c}|h - \psi|$, certifies the closed-loop modulus; (iv) otherwise *freeze* the correction: the blind step, provably stable at $m < 1$, keeps running. The freeze is derived, not tuned. Safety holds on the confidence-sequence event, and certification is not free: the gate opens only after design energy accumulates, which by Theorem 1 is a regret payment (Figure 1).

4 Key results

Every claim above is a preregistered row in a measured-vs-predicted harness on the live pipeline: **36 rows, 0 failures** (35 pass, one deliberately out-of-scope drift cell measured, not excused). Eleven pivots were forced by measurement and recorded in code; three became results (the feedback floor, Theorem 3’s reach, and its below-reach witness), and the derivation suite’s falsification record supplies a fourth, the false h_{PO} -anchored identity (Table 1, Appendix D).

Baselines. Against finite-difference PerfGD [Izzo et al., 2021], zeroth-order P&L optimization, a performative-confidence-bounds explorer [Jagadeesan et al., 2022], and blind RRM on identical environments, none Pareto-dominates SafeD-PerfGD on median (final error, regret) over 12 seeds; the unsafe gradient baseline pays $> 5 \times$ its median regret. We deliberately report: the UCB explorer attains *lower raw regret* and edges SafeD in one of the twelve seeds; certification is a real cost, and its visibility is the thesis. **Second economy.** Transplanted unchanged to LQ performative pricing (deployed price anchors demand), the exchange rate is exact in all four cells (A1 is global in LQ, so the drift vanishes) and the agent reaches the pricing optimum (12.65 vs. 12.5, tail-averaged) once its probes were randomized (alternating probes make price and lag collinear: T6 in the wild). **Real data.** On Nagarajan and Ashok [2026]’s public-data corporate-bond calibration (10 rating $\times$ regime cells), the port matches the published REFLEX run in **10/10** cells ($h^*, \epsilon^* = \gamma$, m within 0.8%; validation inputs live in the REFLEX repository), then extends it: γ_{PO} spans 510.6 to 2.3; echo-chamber gaps are 1.9–4.3% of the anchor spread; and the shaping dispersion is $F = 1.63$ across the portfolio against 1.002 within one book: the shaping gain lives across ratings and regimes, where a desk allocates exploration.

5 Conclusion

An identity, a floor, a design rule, and an agent price the cost of learning one’s own influence: the exchange-rate invariant holds across 36 rows, 0 failures; the floor holds within reach (1.0005) and fails just beyond it (0.8517); and the port matches 10/10 cells within 0.8%.

Limitations and outlook. The identity is local (A1): wide-side probing undercuts it at finite amplitude and the saturating cell drifts +16%, so the $o(1)$ qualifier is load-bearing. So is the trust region: beyond the reach the floor provably fails (Theorem 3(ii)). The value-budget floor is proved to within $13.5 \times$; the sharp constant $\frac{1}{2}$ is numerically supported, not proved, at any prior class. At matched transit energy full-support jitter identifies on par with the design (advantage: small-amplitude, allocational); calibrated moduli are deeply stable and the toxic channel structurally scaled: the real-data leg validates machinery, not effect size.

References

- Alexander Barzykin, Philippe Bergault, Olivier Guéant, and Malo Lemmel. Optimal quoting under adverse selection and price reading. *arXiv preprint arXiv:2508.20225*, 2026.
- Xavier Bombois, Gérard Scorletti, Michel Gevers, Paul M. J. Van den Hof, and Roland Hildebrand. Least costly identification experiment for control. *Automatica*, 42(10):1651–1662, 2006.
- Daniele Bracale, Subha Maity, Yuekai Sun, and Moulinath Banerjee. Learning the distribution map in reverse causal performative prediction. In *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 258 of *PMLR*, 2025.
- Josef Broder and Paat Rusmevichientong. Dynamic pricing under a general parametric choice model. *Operations Research*, 60(4):965–980, 2012.
- Gavin Brown, Shlomi Hod, and Iden Kalemaj. Performative prediction in a stateful world. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2022.
- Sarah Dean, Horia Mania, Nikolai Matni, Benjamin Recht, and Stephen Tu. On the sample complexity of the linear quadratic regulator. *Foundations of Computational Mathematics*, 20(4):633–679, 2020.
- Arnoud V. den Boer. Dynamic pricing and learning: Historical origins, current research, and new directions. *Surveys in Operations Research and Management Science*, 20(1):1–18, 2015.
- Dmitriy Drusvyatskiy and Lin Xiao. Stochastic optimization with decision-dependent distributions. *Mathematics of Operations Research*, 48(2):954–998, 2023.
- A. A. Feldbaum. Dual control theory I. *Automation and Remote Control*, 21:874–880, 1960.
- Richard D. Gill and Boris Y. Levit. Applications of the van Trees inequality: A Bayesian Cramér–Rao bound. *Bernoulli*, 1(1–2):59–79, 1995.
- Olivier Guéant, Charles-Albert Lehalle, and Joaquin Fernandez-Tapia. Dealing with the inventory risk: A solution to the market making problem. *arXiv preprint arXiv:1105.3115*, 2013.
- Moritz Hardt and Celestine Mendler-Dünner. Performative prediction: Past and future. *arXiv preprint arXiv:2310.16608*, 2023.
- J. Michael Harrison, N. Bora Keskin, and Assaf Zeevi. Bayesian dynamic pricing policies: Learning and earning under a binary prior distribution. *Management Science*, 58(3):570–586, 2012.
- Zachary Izzo, Lexing Ying, and James Zou. How to learn when data reacts to your model: Performative gradient descent. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021.
- Meena Jagadeesan, Tijana Zrnic, and Celestine Mendler-Dünner. Regret minimization with performative feedback. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, 2022.
- N. Bora Keskin and Assaf Zeevi. Dynamic pricing with an unknown demand model: Asymptotically optimal semi-myopic policies. *Operations Research*, 62(5):1142–1167, 2014.
- Jack Kiefer and Jacob Wolfowitz. The equivalence of two extremum problems. *Canadian Journal of Mathematics*, 12:363–366, 1960.
- Tze Leung Lai and Herbert Robbins. Adaptive design and stochastic approximation. *The Annals of Statistics*, 7(6):1196–1221, 1979.
- Qiang Li and Hoi-To Wai. State dependent performative prediction with stochastic approximation. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2022.
- Licong Lin and Tijana Zrnic. Plug-in performative optimization. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.

- 223 Celestine Mendler-Dünnér, Juan C. Perdomo, Tijana Zrnic, and Moritz Hardt. Stochastic optimization
224 for performative prediction. In *Advances in Neural Information Processing Systems 33 (NeurIPS)*,
225 2020.
- 226 John Miller, Juan C. Perdomo, and Tijana Zrnic. Outside the echo chamber: Optimizing the
227 performative risk. In *Proceedings of the 38th International Conference on Machine Learning*
228 (*ICML*), 2021.
- 229 Vignesh Nagarajan and Shriraghav Ashok. REFLEX: Reflexive equilibrium fixed-point learning for
230 endogenous exchanges. *arXiv preprint arXiv:2608.16155*, 2026.
- 231 Juan C. Perdomo, Tijana Zrnic, Celestine Mendler-Dünnér, and Moritz Hardt. Performative prediction.
232 In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020.
- 233 Friedrich Pukelsheim. *Optimal Design of Experiments*. SIAM Classics in Applied Mathematics,
234 2006.
- 235 Michael Rothschild. A two-armed bandit theory of market pricing. *Journal of Economic Theory*, 9
236 (2):185–202, 1974.
- 237 Max Simchowitz and Dylan J. Foster. Naive exploration is optimal for online LQR. In *Proceedings*
238 *of the 37th International Conference on Machine Learning (ICML)*, 2020.
- 239 Andrew Wagenmaker, Max Simchowitz, and Kevin Jamieson. Task-optimal exploration in linear
240 dynamical systems. In *Proceedings of the 38th International Conference on Machine Learning*
241 (*ICML*), 2021.

A Theorem statements (register copy)

Lemma 1 (Cost equivalence; L1). *Under A1–A3 with conditionally symmetric exploration, the expected incremental performative cost satisfies $C_T = \frac{1}{2}\gamma_{\text{PO}} \mathbb{E}[\sum_{t \leq T} d_t^2] + R_T$ with $|R_T| \leq \frac{L_3}{6} \mathbb{E} \sum_t |d_t|^3$, and the realized cost’s variance decomposes exactly as $\text{Var}(C_T^{\text{real}}) = \delta_0^2 \text{Var}(\sum_t d_t) + \frac{\gamma_{\text{PO}}^2}{4} \text{Var}(\sum_t d_t^2)$ with vanishing cross term.*

Theorem 4 (Information saturation; T1). *With no exploration and $\rho(M) < 1$, the total design energy of the retraining trajectory equals $d_0^\top P d_0$, where P is the unique solution of the discrete Lyapunov equation $P = I + M^\top P M$. Consequently the Fisher information for the response Jacobian is bounded, in every direction, uniformly in the horizon: $\text{Var}(\hat{\varepsilon}) \geq \sigma^2 (d_0^\top P_u d_0)^{-1}$ for the component along any direction u .*

Corollary 1 (Safety implies blindness; C1.1). *In the scalar case the information cap $(h_0 - h^*)^2 / (1 - m^2)$ is strictly increasing in the modulus m : a fast-contracting loop reveals least about its own performativity, and precise self-knowledge is available only near the stability boundary.*

Corollary 2 (Trajectory design degeneracy; C1.2). *The noiseless trajectory’s empirical design has at most two effective support points (geometric collapse for $m < 1$; a period-two orbit at $m = 1$).*

Remark 1 (Non-normal transient information; R1). *The energy identity $d_0^\top P d_0$ of T1 holds for non-normal M as well, where the energy can exceed the spectral cap $\|d_0\|^2 / (1 - \rho(M)^2)$ by up to the factor $\kappa(V)^2$.*

Theorem 5 (The exchange rate, pathwise; T2). *In the stationary regime, on every realization,*

$$\text{Var}(\hat{\varepsilon} \mid \text{design}) \times \frac{1}{2}\gamma_{\text{PO}} \sum_{t \leq T} d_t^2 = \frac{1}{2}\gamma_{\text{PO}} \sigma^2 \left(1 + \frac{T d^2}{S_{xx}}\right) = \frac{1}{2}\gamma_{\text{PO}} \sigma^2 (1 + O_P(1/T)),$$

independent of the exploration intensity, the horizon, and the modulus.

Proposition 1 (Concentration; P2.1). *In the stationary regime the measured product concentrates around the invariant at rate $T^{-1/2}$, with long-run $\text{Var}(\sum_{t \leq T} d_t) = T v \frac{1-m}{1+m} (1 + o(1))$.*

Proposition 2 (Feedback bias; P2.2). *Under the feedback relaxation A3’ with gain ϕ , the OLS response slope carries the bias*

$$\mathbb{E}[\hat{\varepsilon} - \varepsilon_{\text{true}}] = \frac{\phi \sigma^2 (3m - 1)}{(1 - m^2) T v_\phi} (1 + o(1)), \quad v_\phi = \frac{\sigma_e^2 + \phi^2 \sigma^2}{1 - m^2},$$

which is $O(1/T)$ and changes sign at $m = 1/3$.

Theorem 6 (Minimax lower bound, deviation budget; T3). *Over all non-anticipating exploration policies with $\sum_{t \leq T} d_t^2 \leq D$ almost surely and all estimators (biased included), the minimax risk over an interval of width w satisfies*

$$\inf_{\text{policy, estimator}} \sup_{\varepsilon} \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq \frac{\sigma^2}{D + \sigma^2(\pi/w)^2},$$

so the exchange-rate product is a floor, attained with equality by symmetric probing with least squares.

Lemma 2 (Exploitation–information; L2). *For the symmetric two-point response family with half-width δ and any non-anticipating policy under the certainty-equivalent anchor,*

$$\mathbb{E}[\text{gain}_T] \leq \frac{g_1 \delta^{3/2} T^{1/2} (\mathbb{E} S_T)^{3/4}}{\sigma^{1/2}}, \quad g_1 = \beta |h^* - \psi|,$$

via the exact identity $\mathbb{E}|\mathbb{E}[s \mid \mathcal{F}_t]| = \text{TV}(P_t^+, P_t^-)$ and Pinsker’s inequality $\text{TV}_t \leq \delta \sqrt{\mathbb{E} S_t} / \sigma$: exploitation is bounded by the information already purchased. At the minimax scale $\delta = c \sigma / \sqrt{\mathbb{E} S_T}$ this is $O(\sigma \sqrt{T})$, an $O(T^{-1/2})$ fraction of the $\Theta(T)$ exploration cost.

Theorem 7 (Minimax lower bound, value budget; T4). *Under stationary-scale exploration and the certainty-equivalent anchor, for every non-anticipating policy and every estimator,*

$$\sup \text{Var}(\hat{\varepsilon}) \times C_T \geq \frac{\gamma_{\text{PO}} \sigma^2}{27} (1 - O(T^{-1/2})),$$

the supremum over the two-point response family at the minimax scale and the constant arising from the Le Cam reduction optimized at $c = 2/3$. The exploited rebate vanishes at rate $T^{-1/2}$ (Lemma L2). The sharp constant $\frac{1}{2}$, against which no simulated policy fell by more than Monte-Carlo error (`run_open1.py` §B), is conjectured and is part of OPEN-1.

Theorem 8 (Optimal exploration shapes; T5). *Under the value budget $\text{tr}(\Gamma_{\text{PO}} M) \leq B$ on the exploration second moment M : (a) the A-optimal design is $M^* = (B / \text{tr} \Gamma_{\text{PO}}^{1/2}) \Gamma_{\text{PO}}^{-1/2}$ with value $(\text{tr} \Gamma_{\text{PO}}^{1/2})^2 / B$; (b) the D-optimal design is $M^* = (B/d) \Gamma_{\text{PO}}^{-1}$; (c) the c-optimal design for the correction functional c is $M^* = B c c^\top / (c^\top \Gamma_{\text{PO}} c)$ (probing along the correction direction itself), with value $c^\top \Gamma_{\text{PO}} c / B$.*

Corollary 3 (Price of isotropic jitter; C5.1). *Isotropic exploration at matched budget overpays the A-optimal design by exactly the curvature-dispersion factor $F = d \text{tr}(\Gamma_{\text{PO}}) / (\text{tr} \Gamma_{\text{PO}}^{1/2})^2 \geq 1$, with equality iff all curvature directions are equal.*

Lemma 3 (Temporal shaping is irrelevant; L3). *With serially independent response noise, the information about the response parameters depends on the exploration process only through its empirical second moment; temporally correlated exploration buys nothing beyond its stationary design measure.*

Theorem 9 (Chebyshev unidentifiability; T6). *The sensitivity system of the structural family $\tau(h) = C_0 + C_1 e^{-ch}$ is an extended Chebyshev system; any design with fewer than three support points has singular Fisher information, and by C1.2 no retraining trajectory, at any modulus, identifies the response curvature c .*

Lemma 4 (Perturbed modulus; L4). *If the corrected update uses an estimated response with C^1 error e , the closed-loop modulus satisfies $|\hat{\rho} - \rho| \leq \eta \beta (\|e\|_\infty + |h^* - \psi| \|e'\|_\infty)$.*

Remark 2 (Open-loop neutrality; R2). *History-independent exploration schedules cannot change the linearized contraction rate of the retraining loop.*

Proposition 3 (Safety under pessimism; P7.1). *Given an anytime-valid confidence sequence for the response parameters at level $1 - \delta$, the pessimism rule (worst-case modulus over the confidence set kept below $1 - c_{\text{margin}}$, correction frozen otherwise) keeps the realized modulus within the margin for all t simultaneously with probability at least $1 - \delta$.*

Theorem 10 (Anchoring crossover, horizon-dependent; T7). *Under the value budget B and horizon T , the budget-and-horizon-optimal nonparametric (secant) estimator has*

$$\text{MSE}_{\text{np}} = \frac{\gamma_{\text{PO}} \sigma^2}{2B} + \left(\frac{\tau'''(h^*)}{6} \right)^2 \left(\frac{2B}{\gamma_{\text{PO}} T} \right)^2,$$

so anchoring to a structural family with misspecification δ_{mis} wins iff (to leading order) $\delta_{\text{mis}} < |\tau'''(h^)| B / (3 \gamma_{\text{PO}} T)$: anchoring wins at short horizons and tight budgets, the free-form route asymptotically in T .*

Theorem 11 (The ROI of self-knowledge; T8). *Exploring once to accuracy v and correcting forever, discounted at ρ_{disc} , is optimal at $v^* = (\sigma/\kappa) \sqrt{\rho_{\text{disc}}}$, and exploring at all is worthwhile iff*

$$\rho_{\text{disc}} < \rho^* = \frac{(h_{\text{SP}} - h_{\text{PO}})^4}{4 \kappa^2 \sigma^2},$$

in which the curvature γ_{PO} cancels completely.

Proposition 4 (Frontier consistency; P9.1). *Every deterministic design sits exactly on the exchange-rate frontier; in particular the $O(\log T)$ -cost adaptive-design schedules of Lai–Robbins.*

Proposition 5 (Isotropy contrast; P9.2). *Naive isotropic exploration, optimal in isotropic LQR-like settings, overpays by exactly the dispersion factor F in the performative setting, with $F = 1$ iff the curvature spectrum is flat.*

Proposition 6 (Separable multi-bond curvature; P9.3). *In the separable multi-bond market, $(\Gamma_{\text{PO}})_{aa} = \gamma_a + \beta \varepsilon_a (2 + c_t \psi_a - c_t h_a)$ exactly, with the factor-coupled case holding to first order in the off-diagonal response.*

Theorem 12 (Structure-proofness of the floor, within reach; T9). *Fix the structural family $\tau_\theta(h) = C_0 + C_1 e^{-ch}$ ($C_1 > 0$, $c > 0$), the anchor $h^* > 0$, the sensitivity $s(h) = \partial_\theta \tau_\theta(h)$, and the estimand gradient $g = \partial_\theta \varepsilon(h^*) = -s'(h^*)$. For a finitely supported design μ define the normalized Cramér–Rao product*

$$R(\mu) = (g^\top M(\mu)^{-1} g) \int (h - h^*)^2 d\mu, \quad M(\mu) = \int s s^\top d\mu,$$

so that $R(\mu) \geq 1$ states that the family-knowing Cramér–Rao product $\text{Var}(\hat{\varepsilon}) \times C_T$ is at least the exchange-rate floor $\frac{1}{2}\gamma_{\text{PO}}\sigma^2$ under the local-quadratic (A1) cost, for unbiased estimators and, in the local-asymptotic-minimax sense, for all regular estimators. (i) Within reach the floor is structure-proof: if $\text{supp}(\mu) \subset [h^* - 2/c, \infty)$, then either $\varepsilon(h^*)$ is not estimable under μ (the bound is infinite) or $R(\mu) \geq 1$, strictly whenever μ places mass off the two equality points $\{h^*, h^* - 2/c\}$; and $\inf_{\mu} R(\mu) = 1$, approached but not attained by symmetric three-point designs collapsing at h^* . In particular the floor binds on every trust-region design with $r \leq 2/c$ (A4). (ii) The reach is exact: the pointwise domination underlying (i) fails strictly below $h^* - 2/c$, and an explicit four-point design with a single below-reach probe of weight 3×10^{-4} attains $R = 0.8517$ (computer-verified at 50-digit precision; V9.2): outside the reach, knowledge of the family’s form does beat the floor.

B Proofs

Throughout, the local model of Assumptions A1–A6, per the derivation folder’s notation-and-assumptions register: **A1** (local quadratic scope): Φ is C^3 and strongly concave on the operating interval, expansions taken at h^* with third-order remainder bounded by $L_3 = \sup |\Phi'''|$; **A2** (exploration class): all policies are non-anticipating, with A2-sym additionally requiring $\mathbb{E}[u_t | \mathcal{F}_t] = 0$ and A2-adaptive unrestricted, costs measured from the certainty-equivalent anchor; **A3** (noise exogeneity): ζ_t i.i.d., mean zero, variance σ^2 , independent of $\mathcal{F}_t$; **A4** (trust region): $|d_t| \leq r$ pathwise; **A5** (stationary scale): $S_T = \Theta(T)$; **A6** (stable base loop): $m < 1$, equivalently $\rho(M) < 1$. The local dynamics are $d_{t+1} = -m d_t + u_t$ with $d_t = h_t - h^*$ (h^* the certainty-equivalent operating point, equal to h_{SP} for the blind loop), observations $\tau_t = \tau(h^*) - \varepsilon d_t + \zeta_t$, and Φ satisfying $\Phi'(h^*) = \delta_0 \neq 0$ and $-\Phi'' = \gamma_{\text{PO}}$ at the operating point.

A. Cost equivalence and its fluctuations

Proof of Lemma L1 (cost equivalence). Taylor’s theorem with Lagrange remainder gives, for each t ,

$$\Phi(h^*) - \Phi(h_t) = -\delta_0 d_t + \frac{1}{2}\gamma_{\text{PO}} d_t^2 + r_t, \quad |r_t| \leq \frac{L_3}{6} |d_t|^3,$$

with $L_3 = \sup |\Phi'''|$ on the operating interval. Summing over $t \leq T$ and taking expectations, it remains to show $\mathbb{E}[\sum_t d_t] = 0$ under A2-sym. The recursion gives $\mathbb{E}[d_{t+1} | \mathcal{F}_t] = -m d_t + \mathbb{E}[u_t | \mathcal{F}_t] = -m d_t$, so $\mathbb{E}[d_{t+1}] = -m \mathbb{E}[d_t]$ by the tower property, and $\mathbb{E}[d_t] = (-m)^t \mathbb{E}[d_0] = 0$ when the loop starts at the operating point or in the stationary regime. The quadratic term contributes $\frac{1}{2}\gamma_{\text{PO}} \mathbb{E}[\sum_t d_t^2]$, which is the claim.

For the variance decomposition, write the realized cost (net of the cubic remainder) as $C = -\delta_0 X + \frac{1}{2}\gamma_{\text{PO}} Y$ with $X = \sum_t d_t$, $Y = \sum_t d_t^2$. Then $\text{Var}(C) = \delta_0^2 \text{Var}(X) + \frac{\gamma_{\text{PO}}^2}{4} \text{Var}(Y) - \delta_0 \gamma_{\text{PO}} \text{Cov}(X, Y)$. Under Gaussian innovations the vector $(d_1, \dots, d_T)$ is jointly Gaussian with mean zero, and every third moment of a zero-mean Gaussian vector vanishes; since $\text{Cov}(X, Y) = \sum_{t,s} \mathbb{E}[d_t d_s^2]$ is a sum of third moments, it is zero. (For non-Gaussian symmetric innovations the same conclusion follows from the sign symmetry $d \mapsto -d$ of the stationary law.) Verified: V0.1–V0.3, including the necessity check that an asymmetric rule breaks the expectation identity by exactly $-\delta_0 \mathbb{E}[\sum_t d_t]$. $\square$

B. Saturation

Proof of Theorem T1 (information saturation). Let $\rho(M) < 1$ and define $P = \sum_{t \geq 0} (M^\top)^t M^t$. The series converges absolutely: by Gelfand’s formula $\|M^t\| \leq C \rho_0^t$ for any $\rho(M) < \rho_0 < 1$ and some C , so the terms are dominated by $C^2 \rho_0^{2t}$. Termwise, $M^\top P M = \sum_{t \geq 0} (M^\top)^{t+1} M^{t+1} = P - I$, so P solves the discrete Lyapunov equation, and if Q is any other solution then $P - Q = M^\top (P - Q) M = (M^\top)^k (P - Q) M^k \rightarrow 0$, giving uniqueness. The energy identity is immediate: $\sum_{t \geq 0} \|M^t d_0\|^2 = d_0^\top (\sum_t (M^\top)^t M^t) d_0 = d_0^\top P d_0$. The directional version replaces I by $uu^\top$: $P_u = \sum_t (M^\top)^t uu^\top M^t$ solves $P_u = uu^\top + M^\top P_u M$ and $\sum_t (u^\top M^t d_0)^2 = d_0^\top P_u d_0$.

For the information claim: under A3 the log-likelihood of the observations along the noiseless trajectory has Fisher information for the response component along u equal to $\sigma^{-2} \sum_t (u^\top d_t)^2 = \sigma^{-2} d_0^\top P_u d_0$, which is finite and independent of the horizon; the Cramér–Rao bound gives the

373 stated variance floor for unbiased estimators (the biased/minimax version is Theorem T3's van Trees
374 argument). Verified: V1.1, V1.2, V1.4. $\square$

375 *Proof of Corollary C1.1 (safety implies blindness).* In the scalar case $P = \sum_t m^{2t} = (1 - m^2)^{-1}$,
376 so the cap $(h_0 - h^*)^2/(1 - m^2)$ has derivative in m equal to $2m(h_0 - h^*)^2/(1 - m^2)^2 > 0$ for
377 $m \in (0, 1)$: strictly increasing. $\square$

378 *Proof of Corollary C1.2 (design degeneracy).* For $m < 1$, $d_t = (-m)^t d_0$, so for every $\epsilon > 0$ all
379 but $\lceil \log(|d_0|/\epsilon)/\log(1/m) \rceil$ of the points lie in $[-\epsilon, \epsilon]$: the empirical design measure converges
380 weakly to δ_0 . At $m = 1$, $d_t = (-1)^t d_0$ and the design measure is $\frac{1}{2}(\delta_{+d_0} + \delta_{-d_0})$ exactly. In
381 both cases the limiting design has at most two support points; combined with Theorem T6, the
382 per-observation Fisher matrix of the three-parameter structural family is asymptotically singular
383 along the trajectory. $\square$

384 *Proof of Remark R1 (non-normal transient information).* The identity $E(d_0) = d_0^\top P d_0$ of T1 holds
385 for any stable M , normal or not. If $M = V\Lambda V^{-1}$ is diagonalizable, $\|M^t\| \leq \kappa(V)\rho(M)^t$
386 gives $\|P\| \leq \kappa(V)^2/(1 - \rho(M)^2)$. The bound is attained in order by the Jordan-type family
387 $M = \begin{pmatrix} m & a \\ 0 & m \end{pmatrix}$ with $d_0 = e_2$: $M^t e_2 = (t a m^{t-1}, m^t)^\top$, so

$$E(e_2) = a^2 \sum_{t \geq 1} t^2 m^{2t-2} + \sum_{t \geq 0} m^{2t} = a^2 \frac{1 + m^2}{(1 - m^2)^3} + \frac{1}{1 - m^2},$$

388 using $\sum_{t \geq 1} t^2 x^{t-1} = (1 + x)/(1 - x)^3$ at $x = m^2$: the energy grows as a^2 , without bound, at fixed
389 spectral radius. Verified: V1.3 (the exact value matches the Lyapunov solve; 1268.2 vs the normal
390 cap 2.78 at $m = 0.8$, $a = 6$). $\square$

391 C. The exchange rate

392 *Proof of Theorem T2 (pathwise exchange rate).* Under A3, conditionally on the design $\{d_t\}$ the ob-
393 servations are independent with means linear in ε and variance σ^2 , so the least-squares slope has
394 conditional variance $\text{Var}(\hat{\varepsilon} \mid \text{design}) = \sigma^2/S_{xx}$ (Gauss–Markov). Multiplying by the quadratic cost
395 $\frac{1}{2}\gamma_{\text{PO}} \sum_t d_t^2$ and using $\sum_t d_t^2 = S_{xx} + T\bar{d}^2$ gives the displayed identity exactly, on every realization.
396 In the stationary regime $\bar{d} = O_P(T^{-1/2})$ by Proposition P2.1's long-run variance, while $S_{xx}/T \rightarrow v$
397 a.s., so $T\bar{d}^2/S_{xx} = O_P(1/T)$. Verified: V2.1 (machine precision). $\square$

398 *Proof of Proposition P2.1 (concentration).* For the stationary AR(1) with lag-one coefficient $\rho =$
399 $-m$, $\text{Cov}(d_t, d_s) = v(-m)^{|t-s|}$, so

$$\text{Var}\left(\sum_{t \leq T} d_t\right) = \sum_{t,s} v(-m)^{|t-s|} = Tv \frac{1+\rho}{1-\rho} (1 + o(1)) = Tv \frac{1-m}{1+m} (1 + o(1)),$$

400 the negative autocorrelation of the cobweb shrinking the fluctuation below the i.i.d. level. The realized-
401 cost product then fluctuates around the constant with standard deviation $O(T^{-1/2})$ by Lemma L1's
402 variance decomposition (both $\text{Var}(\sum d)$ and $\text{Var}(\sum d^2)$ are $O(T)$ while the squared mean of the
403 cost is $O(T^2)$). Verified: V2.2 (variance ratio 4.80 for a $4 \times$ horizon), V2.3. $\square$

404 *Proof of Proposition P2.2 (feedback bias).* Under A3', $d_{t+1} = -m d_t + \sigma_e \xi_t + \phi \zeta_t$, so d_t is a
405 linear function of past innovations with $\mathbb{E}[d_s \zeta_t] = \phi \sigma^2 (-m)^{s-t-1}$ for $s > t$ and 0 otherwise.
406 Write the slope error as N/S with $N = \sum_t (d_t - \bar{d}) \zeta_t$ and $S = S_{xx}$, and expand $\mathbb{E}[N/S] =$
407 $\mathbb{E}[N]/\mathbb{E}[S] - \text{Cov}(N, S)/\mathbb{E}[S]^2 + O(T^{-2})$ (valid for stationary Gaussian data; the neglected term
408 involves $\mathbb{E}[N] \text{Var}(S)/\mathbb{E}[S]^3 = O(T^{-2})$).

409 *Centering term.* $\mathbb{E}[d_t \zeta_t] = 0$, so $\mathbb{E}[N] = -\frac{1}{T} \sum_{s>t} \mathbb{E}[d_s \zeta_t] = -\frac{1}{T} \sum_{s>t} \phi \sigma^2 (-m)^{s-t-1} =$
410 $-\phi \sigma^2/(1 + m) + O(1/T)$.

411 *Covariance term.* For jointly Gaussian zero-mean (X, Y, Z) , $\mathbb{E}[XY^2Z] = 2\mathbb{E}[XY]\mathbb{E}[YZ] +$
 412 $\mathbb{E}[XZ]\mathbb{E}[Y^2]$ (Isserlis). Applying this to $\text{Cov}(N, S) = \sum_{t,s} (\mathbb{E}[\tilde{d}_t \tilde{d}_s^2 \zeta_t] - \mathbb{E}[\tilde{d}_t \zeta_t] \mathbb{E}[\tilde{d}_s^2])$ with
 413 $\tilde{d} = d - \bar{d}$ leaves only the paired term $2 \sum_{t,s} \mathbb{E}[\tilde{d}_t \tilde{d}_s] \mathbb{E}[\tilde{d}_s \zeta_t]$; substituting the stationary covariances,

$$\text{Cov}(N, S) = 2T v_\phi \phi \sigma^2 \sum_{k \geq 1} (-m)^k (-m)^{k-1} + O(1) = -\frac{2T v_\phi \phi \sigma^2 m}{1 - m^2} + O(1).$$

414 Combining with $\mathbb{E}[S] = T v_\phi (1 + o(1))$:

$$\mathbb{E}[\hat{\varepsilon} - \varepsilon_{\text{true}}] = \frac{\phi \sigma^2}{T v_\phi} \left[-\frac{1}{1+m} + \frac{2m}{1-m^2} \right] + O(T^{-2}) = \frac{\phi \sigma^2 (3m-1)}{(1-m^2) T v_\phi} + O(T^{-2}).$$

415 The bracket changes sign at $m = 1/3$. (The first draft kept only the centering term and was falsified
 416 by the verification suite; the record is in VERIFICATION.md.) Verified: V2.4, V2.5 (both signs,
 417 straddling $m = 1/3$). $\square$

418 D. Minimax lower bounds

419 *Proof of Theorem T3 (deviation-budget minimax).* Let π be a prior density on an interval of width w
 420 with finite information $I(\pi) = \int (\pi')^2 / \pi$; the cosine-squared density attains the minimum $I(\pi) =$
 421 $(\pi/w)^2$ (the latter π is the constant). For any non-anticipating policy the joint density of the
 422 data factorizes as $\prod_t p(u_t | \mathcal{F}_t) p(\tau_t | \mathcal{F}_t, \varepsilon)$, where the policy factors do not depend on ε and
 423 $\tau_t | \mathcal{F}_t \sim \mathcal{N}(\tau(h^*) - \varepsilon d_t, \sigma^2)$ with d_t being $\mathcal{F}_t$ -measurable. Hence the Fisher information of the
 424 whole experiment is

$$I_T(\varepsilon) = \mathbb{E}_\varepsilon \left[\sum_t (\partial_\varepsilon \log p(\tau_t | \mathcal{F}_t, \varepsilon))^2 \right] = \mathbb{E}_\varepsilon \left[\sum_t d_t^2 \right] / \sigma^2 \leq D / \sigma^2$$

425 under the pathwise budget: adaptivity places information but cannot manufacture it. The van Trees
 426 inequality (Gill–Levit) then bounds the Bayes risk of *any* estimator, biased or not: $\mathbb{E}_\pi \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq$
 427 $(\mathbb{E}_\pi[I_T] + I(\pi))^{-1} \geq (D/\sigma^2 + (\pi/w)^2)^{-1}$, and the supremum over the interval dominates the prior
 428 average. Equality (up to the prior term) is attained by symmetric two-point probing at the budget
 429 with least squares, by Theorem T2. Verified: V3.1, V3.2 (equality within Monte-Carlo error). $\square$

430 *Proof of Lemma L2 (exploitation–information).* Prior $s = \pm 1$ symmetric; environments $P^\pm$ differ-
 431 ing only in the observation means by $2\delta d_t$ at step t . Three ingredients.

432 (i) *Posterior imbalance equals total variation in expectation.* Under the symmetric mixture $\bar{P} =$
 433 $\frac{1}{2}(P^+ + P^-)$, with data x ,

$$\mathbb{E}_{\bar{P}} |\mathbb{P}(s=+ | x) - \mathbb{P}(s=- | x)| = \int \frac{1}{2}(dP^+ + dP^-) \frac{|dP^+ - dP^-|}{dP^+ + dP^-} = \text{TV}(P_t^+, P_t^-).$$

434 (ii) *Pinsker with the chain rule for adaptive designs.* $\text{KL}(P_t^+ \| P_t^-) = \mathbb{E}^+ [\sum_{i \leq t} (2\delta d_i)^2 / (2\sigma^2)] =$
 435 $2\delta^2 \mathbb{E}^+[S_t] / \sigma^2$ (the policy factors cancel inside the log-ratio), so $\text{TV}_t \leq \delta \sqrt{\mathbb{E}[S_t]} / \sigma$.

436 (iii) *The gain bound.* Since d_t is $\mathcal{F}_t$ -measurable, $\mathbb{E}[s d_t] = \mathbb{E}[d_t \mathbb{E}[s | \mathcal{F}_t]]$; two applications of
 437 Cauchy–Schwarz and $|\mathbb{E}[s | \mathcal{F}_t]| \leq 1$ give

$$\mathbb{E}[\text{gain}_T] = g_1 \delta \sum_t \mathbb{E}[s d_t] \leq g_1 \delta \sum_t (\mathbb{E} d_t^2)^{1/2} \text{TV}_t^{1/2} \leq g_1 \delta (\mathbb{E} S_T)^{1/2} (T \text{TV}_T)^{1/2},$$

438 and inserting (ii): $\mathbb{E}[\text{gain}_T] \leq g_1 \delta^{3/2} T^{1/2} (\mathbb{E} S_T)^{3/4} / \sigma^{1/2}$.

439 *Remark on constants.* The derivation document’s heuristic chain (treating the imbalance as bounded
 440 by TV_t pathwise) yields $g_1 \delta^2 S_T \sqrt{T} / \sigma$; the rigorous chain above changes the exponent bookkeeping
 441 but not the conclusion of Theorem T4: at the minimax scale $\delta = c \sigma / \sqrt{\mathbb{E} S_T}$ both give $\mathbb{E}[\text{gain}_T] =$
 442 $O(\sigma \sqrt{T})$, hence an exploited fraction $O(T^{-1/2})$ of the $\Theta(T)$ cost under A5. Verified: V3.3 (the
 443 imbalance–TV inequality pathwise in expectation), V3.4 (the $T^{-1/2}$ decay at the minimax scale). $\square$

444 *Proof of Theorem T4 (value-budget minimax; constant-factor version).* Fix a policy and estimator;
 445 let $\bar{S} = \mathbb{E}[S_T] = \Theta(T)$ under A5. Choose the two-point prior at separation $\delta = c\sigma/\sqrt{\bar{S}}$ with
 446 $c \in (0, 1)$ to be optimized. By the testing reduction (Le Cam), any estimator's worst-case risk
 447 over the pair obeys $\sup_{\pm} \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq \frac{\delta^2}{2} (1 - \text{TV}_T) \geq \frac{\delta^2}{2} (1 - c)$ using Lemma L2(ii). By
 448 Lemma L1 with the certainty-equivalent anchor and Lemma L2's gain bound, the value cost obeys
 449 $C_T \geq \frac{1}{2} \gamma_{\text{PO}} \bar{S} (1 - O(T^{-1/2}))$. Multiplying,

$$\sup \text{Var}(\hat{\varepsilon}) \times C_T \geq \frac{c^2 \sigma^2}{2\bar{S}} (1 - c) \cdot \frac{\gamma_{\text{PO}} \bar{S}}{2} (1 - O(T^{-1/2})) = \frac{\gamma_{\text{PO}} \sigma^2}{4} c^2 (1 - c) (1 - O(T^{-1/2})),$$

450 maximized at $c = 2/3$: the bound $\gamma_{\text{PO}} \sigma^2 / 27 \cdot (1 - O(T^{-1/2}))$.

451 *Status.* This proves the value-budget product is bounded below by an explicit constant multiple
 452 of the exchange rate, uniformly over adaptive policies, with the exploited rebate vanishing at rate
 453 $T^{-1/2}$. The sharp constant $\frac{1}{2} \gamma_{\text{PO}} \sigma^2$, numerically supported (run_open1.py §B: three adaptive
 454 amplitude schedules, ratios 0.995 to 1.018), requires the van Trees route with cost control in place of
 455 the two-point reduction and is part of OPEN-1. $\square$

456 E. Design geometry

457 *Proof of Theorem T5 (optimal exploration shapes).* All three problems have convex (resp. concave)
 458 objectives over the PSD cone with one linear constraint; KKT stationarity plus convexity/concavity
 459 certifies global optimality.

460 (a) *A-optimality.* $\nabla_M \text{tr}(M^{-1}) = -M^{-2}$, so stationarity gives $M^{-2} = \lambda \Gamma_{\text{PO}}$, i.e. $M =$
 461 $\lambda^{-1/2} \Gamma_{\text{PO}}^{-1/2}$; the budget $\text{tr}(\Gamma_{\text{PO}} M) = \lambda^{-1/2} \text{tr}(\Gamma_{\text{PO}}^{1/2}) = B$ pins $\lambda^{-1/2} = B / \text{tr}(\Gamma_{\text{PO}}^{1/2})$
 462 and the value $\text{tr}(M^{-1}) = (\text{tr} \Gamma_{\text{PO}}^{1/2})^2 / B$. (Diagonal-case cross-check by Cauchy–Schwarz:
 463 $(\sum_i 1/m_i)(\sum_i g_i m_i) \geq (\sum_i \sqrt{g_i})^2$ with equality at $m_i \propto g_i^{-1/2}$.)

464 (b) *D-optimality.* $\nabla_M \log \det M = M^{-1} = \lambda \Gamma_{\text{PO}}$ gives $M = (B/d) \Gamma_{\text{PO}}^{-1}$ after normalizing by the
 465 budget.

466 (c) *c-optimality.* Substitute $N = \Gamma_{\text{PO}}^{1/2} M \Gamma_{\text{PO}}^{1/2}$ ($\text{tr} N = \text{tr}(\Gamma_{\text{PO}} M) \leq B$) and $w = \Gamma_{\text{PO}}^{1/2} c$,
 467 so the objective is $w^\top N^{-1} w$. By Cauchy–Schwarz, $(w^\top w)^2 = (w^\top N^{-1/2} N^{1/2} w)^2 \leq$
 468 $(w^\top N^{-1} w)(w^\top N w)$ and $w^\top N w \leq \|w\|^2 \text{tr} N \leq \|w\|^2 B$, hence $w^\top N^{-1} w \geq \|w\|^2 / B =$
 469 $c^\top \Gamma_{\text{PO}} c / B$, attained at $N^* = B w w^\top / \|w\|^2$. Back-substituting with $\Gamma_{\text{PO}}^{-1/2} w = c$ gives
 470 $M^* = B c c^\top / (c^\top \Gamma_{\text{PO}} c)$: the optimal probe direction is c itself, the curvature pricing only its
 471 scale. Verified: V4.1–V4.3 (formulas achieved; never beaten by 4000 random feasible designs
 472 each). $\square$

473 *Proof of Corollary C5.1 (price of isotropic jitter).* Isotropic $M = (B / \text{tr} \Gamma_{\text{PO}}) I$ has $\text{tr}(M^{-1}) =$
 474 $d \text{tr}(\Gamma_{\text{PO}}) / B$; dividing by the optimum $(\text{tr} \Gamma_{\text{PO}}^{1/2})^2 / B$ gives $F = d \text{tr}(\Gamma_{\text{PO}}) / (\text{tr} \Gamma_{\text{PO}}^{1/2})^2$, and $F \geq 1$
 475 with equality iff all eigenvalues are equal, by Cauchy–Schwarz applied to $(\sum_i \sqrt{g_i} \cdot 1)^2 \leq d \sum_i g_i$.
 476 Verified: V4.4. $\square$

477 *Proof of Lemma L3 (temporal shaping).* Under A3 the conditional log-likelihood of the responses
 478 given the deployment path is $-\sum_t (\tau_t - a - \theta^\top d_t)^2 / (2\sigma^2)$ up to constants; its Hessian in θ is
 479 $-\sum_t d_t d_t^\top / \sigma^2$, a function of the empirical second moment only, whatever the temporal dependence
 480 of $\{d_t\}$. Verified: V4.5. $\square$

481 *Proof of Theorem T6 (Chebyshev unidentifiability).* The sensitivity of the family $\tau(h) = C_0 +$
 482 $C_1 e^{-ch}$ is $s(h) = (1, e^{-ch}, -C_1 h e^{-ch})^\top$. The functions $\{1, e^{-ch}, h e^{-ch}\}$ form an extended
 483 Chebyshev system on any interval (their Wronskian is $c^3 e^{-2ch} \neq 0$ after row reduction; equivalently,
 484 a nontrivial linear combination $a + (b + \gamma h) e^{-ch}$ has at most two zeros counting multiplicity). For
 485 any design ξ with k support points, the Fisher matrix $M(\xi) = \sum_{i \leq k} w_i s(h_i) s(h_i)^\top$ has rank at
 486 most k ; with $k \leq 2 < 3$ it is singular, and that null space always loads on the c -sensitivity: writing n
 487 for the null vector of the two-point design matrix, $n_3 = 0$ would force $n_1 + n_2 e^{-ch_i} = 0$ at both

support points, hence $n = 0$ for $h_1 \neq h_2$, a contradiction. By Corollary C1.2 the trajectory design has at most two effective support points, which proves the claim; the existence of exactly-three-point D-optimal designs on a trust-region interval is classical for Chebyshev systems (Karlin–Studden). Verified: V5.1 (two-point determinants at 10^{-17} ; three-point at 6×10^{-3}). $\square$

492 F. Safety under certainty equivalence

493 *Proof of Lemma L4 (perturbed modulus).* The corrected map is $\Psi(h) = h + \eta[G(h) - \beta(h - \psi)\hat{\varepsilon}(h)]$ and the ideal map replaces $\hat{\varepsilon}$ by ε . With $e = \hat{\varepsilon} - \varepsilon$, $\Psi'(h) - \Psi'_{\text{ideal}}(h) = -\eta\beta \frac{d}{dh}[(h - \psi)e(h)] = -\eta\beta[e(h) + (h - \psi)e'(h)]$, and taking suprema over the operating interval gives the bound with $|h - \psi|_{\max}$; reading the linearized modulus at the operating point h^* (where the certificate is evaluated) localizes the constant to the registered $|h^* - \psi|$. Verified: V6.1 (nine-cell error grid; the bound holds with maximal excess 2.4×10^{-11}). $\square$

499 *Proof of Remark R2 (open-loop neutrality).* For $d_{t+1} = -m d_t + u_t$ with a history-independent schedule $\{u_t\}$, two solutions with different initial conditions differ by the homogeneous solution $(-m)^t(d_0 - d'_0)$, which contracts at rate m regardless of the schedule (superposition). Verified: V6.2 (difference exactly 0 after 200 steps to machine precision). $\square$

503 *Proof of Proposition P7.1 (safety under pessimism).* Let E be the event that the confidence sequence contains the true response parameters simultaneously for all t ; by anytime-validity $\mathbb{P}(E) \geq 1 - \delta$. On E , the true parameters lie in Θ_t at every step, so the realized modulus is dominated by the rule's worst case: $|\rho_t| \leq \sup_{\theta \in \Theta_t} |\hat{\rho}(\theta)| \leq 1 - c_{\text{margin}}$ whenever correction is active, and when the rule freezes, the blind loop's modulus $m < 1 - c_{\text{margin}}$ (choose the margin below $1 - m$) applies. No union bound over time is needed: the inclusion is pathwise on E . (Conditional result: the hypothesis is the confidence sequence's validity; constructing it from the D2 information accounting is routine, and the necessity of the hypothesis is demonstrated numerically: an invalid, too-narrow sequence produces violations.) $\square$

512 G. The anchoring crossover

513 *Proof of Theorem T7 (horizon-dependent crossover).* Bias. For the symmetric secant with half-width w , fifth-order Taylor expansion of $\tau(h^* \pm w)$ gives

$$\frac{\tau(h^* + w) - \tau(h^* - w)}{2w} = \tau'(h^*) + \frac{\tau'''(h^*)}{6} w^2 + O(w^4),$$

515 all even-order terms cancelling (symmetric probing is bias-optimal at its order; the second derivative never enters). Verified: V7.1.

517 *Variance.* With n probe pairs ($T = 2n$ observations) the slope estimator $(\bar{y}_+ - \bar{y}_-)/(2w)$ has variance $\frac{2\sigma^2/n}{4w^2} = \frac{\sigma^2}{2nw^2} = \frac{\sigma^2}{S}$ with $S = \sum_t d_t^2 = 2nw^2 = Tw^2$: variance depends on the design only through its energy.

520 *Constrained optimum.* The value budget fixes $S = 2B/\gamma_{\text{PO}}$; the horizon caps the number of observations, $Tw^2 \geq S$, i.e. $w^2 \geq 2B/(\gamma_{\text{PO}}T)$; bias is increasing in w , so the constrained optimum sits at $w^2 = 2B/(\gamma_{\text{PO}}T)$, giving

$$\text{MSE}_{\text{np}} = \frac{\gamma_{\text{PO}} \sigma^2}{2B} + \left(\frac{\tau'''(h^*)}{6} \right)^2 \left(\frac{2B}{\gamma_{\text{PO}}T} \right)^2.$$

523 Verified: V7.2–V7.4 (three (B, T) cells, within 2%).

524 *Crossover.* The anchored estimator has $\text{MSE} = \kappa_p \gamma_{\text{PO}} \sigma^2 / (2B) + \delta_{\text{mis}}^2$; comparing and keeping the dominant term gives the rule $\delta_{\text{mis}} < |\tau'''(h^*)| B / (3\gamma_{\text{PO}}T)$ to leading order, with the exact two-term form in the display above. The direction of the comparison on both sides of the threshold is verified: V7.5, V7.6. $\square$

528 H. ROI and instantiation

529 *Proof of Theorem T8 (ROI of self-knowledge).* With $A = \frac{1}{2}\gamma_{\text{PO}}(h_{\text{SP}} - h_{\text{PO}})^2$, $a = \frac{1}{2}\gamma_{\text{PO}}\kappa^2$, $b =$
530 $\frac{1}{2}\gamma_{\text{PO}}\sigma^2$, the net present value of exploring to accuracy v is $\text{NPV}(v) = (A - av)/\rho_{\text{disc}} - b/v$, strictly
531 concave on $(0, \infty)$ with $\text{NPV}'(v) = -a/\rho_{\text{disc}} + b/v^2$, giving $v^* = \sqrt{b\rho_{\text{disc}}/a} = (\sigma/\kappa)\sqrt{\rho_{\text{disc}}}$
532 (the curvature cancels between a and b). Substituting, $\text{NPV}(v^*) = A/\rho_{\text{disc}} - 2\sqrt{ab/\rho_{\text{disc}}}$, which
533 is positive iff $A > 2\sqrt{ab\rho_{\text{disc}}}$, i.e. iff $\rho_{\text{disc}} < A^2/(4ab) = (h_{\text{SP}} - h_{\text{PO}})^4/(4\kappa^2\sigma^2)$: γ_{PO} cancels
534 completely. Verified: V8.1, V8.2 (the break-even root to 10^{-3} relative). $\square$

535 *Proof of Proposition P9.1 (frontier consistency).* For any deterministic design, $\text{Var}(\hat{\varepsilon} \mid \text{design}) \cdot$
536 $\frac{1}{2}\gamma_{\text{PO}}S = (\sigma^2/S) \cdot \frac{1}{2}\gamma_{\text{PO}}S = \frac{1}{2}\gamma_{\text{PO}}\sigma^2$: the design energy cancels identically, whatever the schedule,
537 including the $d_t \propto t^{-1/2}$ schedules with $S_T = \Theta(\log T)$ of the Lai–Robbins adaptive-design line.
538 Verified: V8.3 (exact). $\square$

539 *Proof of Proposition P9.2 (isotropy contrast).* Immediate from Corollary C5.1: $F = 1$ iff the cur-
540 vature spectrum is flat, the isotropic (LQR-like) case in which naive exploration is optimal; any
541 curvature dispersion makes $F > 1$ by the strict case of Cauchy–Schwarz. Cross-verified with
542 V4.4. $\square$

543 *Proof of Theorem T9 (structure-proofness within reach).* Write $x = h - h^*$, $d(h) = x$, and $V =$
544 $\text{span}\{1, e^{-ch}, he^{-ch}\}$, the span of the components of s (a bijective correspondence $u \leftrightarrow \phi_u = u^\top s$
545 between $\mathbb{R}^3$ and V , since $C_1 \neq 0$). Differentiating s gives $s'(h) = (0, -ce^{-ch}, C_1e^{-ch}(ch - 1))^\top$,
546 so $g = -s'(h^*)$ componentwise, which is the stated estimand gradient.

547 *Step 1: one-dimensional reduction.* For $M(\mu) \succ 0$ the generalized Rayleigh identity gives

$$g^\top M(\mu)^{-1}g = \sup_{u \neq 0} \frac{(u^\top g)^2}{u^\top M(\mu)u} = \sup_{\phi \in V} \frac{\phi'(h^*)^2}{\int \phi^2 d\mu} = \left(\inf \left\{ \int \phi^2 d\mu : \phi \in V, \phi'(h^*) = 1 \right\} \right)^{-1},$$

548 using $u^\top Mu = \int \phi_u^2 d\mu$ and $u^\top g = -\phi'_u(h^*)$ (the sign is lost in the square). Hence

$$R(\mu) = \frac{\int d^2 d\mu}{\inf \left\{ \int \phi^2 d\mu : \phi \in V, \phi'(h^*) = 1 \right\}}.$$

549 *Step 2: the candidate and its pointwise domination.* Let

$$\phi_0(h) = \frac{2}{c} - e^{-cx} \left(\frac{2}{c} + x \right) = \frac{2}{c} \cdot 1 - e^{ch^*} \left(\frac{2}{c} - h^* \right) e^{-ch} - e^{ch^*} (he^{-ch}) \in V,$$

550 the unique element of V with 2-jet $(0, 1, 0)$ at h^* (the 2-jet map is invertible on V : in the basis
551 $\{1, e^{-ch}, he^{-ch}\}$ its determinant is $c^2 e^{-2ch^*} \neq 0$); the expansion is $\phi_0(x) = x - (c^2/6)x^3 + O(x^4)$.
552 Writing $f(x) = \frac{2}{c} - e^{-cx}(\frac{2}{c} + x)$ we claim

$$|f(x)| \leq |x| \quad \text{for all } x \geq -2/c, \quad \text{with equality iff } x \in \{0, -2/c\},$$

553 and $|f(x)| > |x|$ strictly for every $x < -2/c$. Indeed: (a) $v(x) = f(x) - x$ has $v(0) = 0$ and
554 $v'(x) = e^{-cx}(1 + cx) - 1 < 0$ for $x \neq 0$ (since $1 + t < e^t$ for $t = cx \neq 0$); $v' \leq 0$ with equality
555 only at the isolated point 0 makes v strictly decreasing on $\mathbb{R}$, so $f > x$ on $x < 0$ and $f < x$ on
556 $x > 0$. (b) $f(0) = 0$ and $f'(x) = e^{-cx}(1 + cx) > 0$ for $x \geq 0$, so $f \geq 0$ on $[0, \infty)$ and, with (a),
557 $0 \leq f \leq x$ there. (c) $q(x) = f(x) + x$ has $q(-2/c) = 0 = q(0)$ and $q'(x) = e^{-cx}(1 + cx) + 1$,
558 where $r(x) = e^{-cx}(1 + cx)$ is strictly increasing on $x < 0$ ($r'(x) = -c^2 x e^{-cx} > 0$) from
559 $r(-2/c) = -e^2 < -1$ to $r(0) = 1$; so q' changes sign exactly once on $(-2/c, 0)$, from negative
560 to positive, and $q \leq 0$ there with equality only at the endpoints; with (a) ($f \geq x$, equality only
561 at 0) this gives $x \leq f \leq -x$ on $[-2/c, 0]$. (d) For $x < -2/c$, $r(x) < -e^2$ gives $q' < 0$, so
562 moving left from $-2/c$ the function q increases above 0: $f(x) > -x = |x| > 0$, and in fact
563 $f(x) = 2/c + e^{c|x|}(|x| - 2/c)$ grows exponentially.

564 *Step 3: the bound.* If $\text{supp}(\mu) \subset [h^* - 2/c, \infty)$, Step 2 gives $\int \phi_0^2 d\mu \leq \int d^2 d\mu$, so by Step 1,
565 $R(\mu) \geq \int d^2 d\mu / \int \phi_0^2 d\mu \geq 1$. Strictness: equality in Step 2 holds only at $x \in \{0, -2/c\}$, so any
566 μ placing positive mass off $\{h^*, h^* - 2/c\}$ (in particular any with three distinct support points)

has $\int \phi_0^2 d\mu < \int d^2 d\mu$ and $R(\mu) > 1$. If $M(\mu)$ is singular, two cases: $g \notin \text{range } M(\mu)$ makes $\varepsilon(h^*)$ non-estimable (infinite bound); $g \in \text{range } M(\mu)$, which does occur for special two-point designs (a one-parameter family of in-reach pairs; their finite pseudo-inverse products were checked to obey the bound, e.g. $R = 1.063$ at the pair $\{1.627, 2.127\}$), keeps the sup-representation valid over $\{u : u^\top M u > 0\}$, and $\int \phi_0^2 d\mu = 0$ would put $u_0 \in \text{null } M$ with $u_0^\top g = -1 \neq 0$, contradicting $g \in \text{range } M$; so $\int \phi_0^2 d\mu > 0$ and the bound goes through verbatim. Throughout, the Cramér–Rao functional bounds unbiased estimation, and all regular estimators in the local-asymptotic-minimax sense.

Step 4: the infimum is 1. The Cramér–Rao functional is invariant under C^1 reparametrization, and the 2-jet map $\theta \mapsto (\tau_\theta(h^*), \varepsilon(h^*), \tau''_\theta(h^*))$ is a local diffeomorphism: its Jacobian $[s(h^*) \ -s'(h^*) \ s''(h^*)]$ has determinant $C_1 c^2 e^{-2ch^*} \neq 0$ (direct computation; V9.4). In jet coordinates η the model reads $\tau = \eta_1 - \eta_2 x + \frac{1}{2} \eta_3 x^2 + O(x^3)$ with sensitivity $\tilde{s}(x) = (1, -x, x^2/2)^\top + O(x^3)$, and the estimand is η_2 . For the symmetric three-point design μ_δ uniform on $\{h^* - \delta, h^*, h^* + \delta\}$, scale by $D = \text{diag}(1, \delta, \delta^2)$: $M_\eta(\mu_\delta) = D \hat{H}_\delta D$ with $\hat{H}_\delta \rightarrow \hat{H}_0$ invertible (the exact quadratic-model moment matrix), so $(M_\eta^{-1})_{22} = \delta^{-2} (\hat{H}_\delta^{-1})_{22}$ and $R(\mu_\delta) \rightarrow R_{\text{quad}}$. In the exact quadratic model, for any symmetric design (odd moments $m_1 = m_3 = 0$), the $(2, 2)$ cofactor gives $(M_\eta^{-1})_{22} = 1/m_2$ and $R_{\text{quad}} = m_2 \cdot (1/m_2) = 1$. Hence $R(\mu_\delta) \rightarrow 1$ (measured rate $R(\mu_\delta) - 1 = O(\delta^2)$; V9.2), the infimum is 1, and by Step 3 it is not attained.

Step 5: the reach is exact. Below $h^* - 2/c$ the domination of Step 2 fails strictly, and the failure is realized by a design, not only by the candidate: the frozen four-point witness (support $\{1.8726, 1.8665, 1.3073, 0.0500\}$, weights $\{0.003329, 0.955702, 0.040697, 0.000272\}$ after normalization, in the register market $c = 1.5$, frozen anchor $h^* = 1.8461$) has $R = 0.851650 \dots$, verified at 50-digit precision (mpmath) and reproduced in float64 by `run_open1.py` (V9.2). All its support except the probe at 0.05 lies within reach: one vanishing-weight below-reach probe breaks the floor. Symmetric pair-plus-far-probe families do *not* break it (their ratio approaches 1 from above as the probe weight vanishes); the violation requires the asymmetric cluster, which is why coarse searches missed it. Verified: V9.1–V9.4. $\square$

Proof of Proposition P9.3 (separable multi-bond curvature). In the separable case the objective is a sum of per-bond objectives $\Phi_a(h_a)$, each of the scalar form analyzed in the single-book theory of Nagarajan and Ashok [2026], so the Hessian is diagonal and the scalar computation applies per bond: differentiating $\Phi_a(h) = P[h A_a e^{-k_a h} \rho + h \tau_a(h) - \psi_a \tau_a(h) - w_a (h - h_{\text{ref}})^2]$ twice and collecting terms with $\tau'_a = -\varepsilon_a$, $\tau''_a = c_t \varepsilon_a$ yields $-\Phi''_a = \gamma_a + \beta \varepsilon_a (2 + c_t \psi_a - c_t h_a)$, which is the claimed diagonal entry. In the factor-coupled case the off-diagonal response contributes additional terms of first order in $\|E_{\text{offdiag}}\|$, stated as a construction with its error term (not claimed exact). Verified: V8.4 (finite-difference Hessians of three random bonds, relative error below 10^{-4}). $\square$

602 C Complete derivations

This appendix reproduces the full derivation record behind every register result: the model’s closed forms, both budgets, and every intermediate computation, including the measurement-forced corrections recorded where they happened. Sections carry the derivation-record identifiers (D0–D9) that the register and Appendices A–B cite.

607 Notation, assumptions, and the two anchoring conventions

This subsection is the canonical register for the derivation record D0 through D9, with D3b the adaptive companion of D3: D0 is the model and cost lemma, D1 information saturation, D2 the exchange rate, D3 the minimax lower bounds, D4 and D5 the design geometry, D6 safe certainty equivalence, D7 the anchoring crossover, and D8 and D9 the ROI analysis and the instantiation. Every symbol is defined once, here; every assumption is numbered here and is cited by number in the derivation sections and in the register statements (T1, L2, $\dots$). Where a later section restates a condition, this register is authoritative. The formal proofs of the register statements are collected in the proofs appendix.

616 **Notation.** The symbol table is reproduced in three blocks: the retraining loop and its objective, the
617 stochastic and informational quantities, and the symbols specific to the design and economic analyses.
618 The third column records where each symbol is introduced: a derivation section (D0 through D9),
619 the base structural market theory (cited as REFLEX [Nagarajan and Ashok, 2026] with its section
620 numbers, a third-party source), or an entry of the assumption register below (the trust-region radius
621 r is fixed by A4). Two register conventions carry over unchanged: all logarithms are natural, and
622 a prime denotes the derivative of a scalar function (in the ASCII sources it also denotes the matrix
623 transpose; in this appendix transposes are written $(\cdot)^\top$, so the prime is unambiguous).

Symbol	Meaning	Defined in / source
$\bar{h}, h_t$	deployed decision (half-spread); its value at step t	D0
h^*	performatively stable point (RRM fixed point)	REFLEX [Nagarajan and Ashok, 2026]
h_{SP}, h_{PO}	stable point ($h_{SP} = h^*$); performative optimum	REFLEX [Nagarajan and Ashok, 2026]
d_t	deviation $d_t = h_t - h^*$	D0
m	retraining modulus $m = \varepsilon\beta/\gamma \in [0, 1)$	REFLEX [Nagarajan and Ashok, 2026]
624 $\gamma, \beta, \varepsilon, \psi$	curvature, smoothness, response slope, adverse severity	REFLEX [Nagarajan and Ashok, 2026] (closed forms)
Φ	true performative objective $\Phi(h) = J(h; \tau(h))$	REFLEX [Nagarajan and Ashok, 2026]
δ_0	$\Phi'(h^*) = -\beta(h^* - \psi)\varepsilon(h^*)$, nonzero	D0 Sec. 1
γ_{PO}	$-\Phi''$ at the operating point	REFLEX [Nagarajan and Ashok, 2026]
M	d -dimensional loop Jacobian, $\rho(M) < 1$	D1 Sec. 2
P, P_u	discrete-Lyapunov energy matrices	D1 Sec. 2
Γ_{PO}	d -dimensional objective curvature matrix	D9

Symbol	Meaning	Defined in / source
σ	response-noise standard deviation (σ_τ in REFLEX)	D0
σ_e, u_t	exploration intensity; exploration input	D0
ξ_t, ζ_t	deployment and response noise (iid, mean zero)	D0
ϕ	noise-feedback gain (observed noise flowing into the next deployment)	D2 Sec. 4
v	stationary deviation variance $\sigma_e^2/(1 - m^2)$	D2
v_ϕ	the same with feedback: $(\sigma_e^2 + \phi^2\sigma^2)/(1 - m^2)$	D2 Sec. 4
625 S_{xx}	centered design energy $\sum_t (d_t - \bar{d})^2$	D2
D_T	deviation budget $\sum_t d_t^2$	D0 Sec. 2
C_T	value budget: expected incremental cost (L1)	D0 Secs. 2–3
I_T	Fisher information for ε	D1/D3
$\mathcal{F}_t$	history sigma-field $\sigma(d_0, u_{<t}, \tau_{<t})$	D0
δ, s	two-point prior half-width; its sign (minimax)	D3
g_1	$\beta h^* - \psi $, coefficient of the unknown linear part	D3
TV, KL	total variation; Kullback–Leibler divergence	D3
r	trust-region radius on $ d_t $	A4

Symbol	Meaning	Defined in / source
M (design)	exploration second moment $\mathbb{E}[d d^\top]$; appears only in D4	D4
B	value budget in the design problems	D4
F	curvature dispersion $d \operatorname{tr}(\Gamma_{\text{PO}})/(\operatorname{tr} \Gamma_{\text{PO}}^{1/2})^2$	D4 (C5.1)
c	correction-direction vector (c -optimality)	D4 Sec. 3
$s(h)$	structural-family sensitivity vector, $p = 3$	D5
w	probe half-width (crossover analysis)	D7
δ_{mis}	C^1 misspecification of the structural family	D6/D7
κ	correction sensitivity $ dh_{\text{PO}}/d\varepsilon $	D8
ρ_{disc}	discount rate (ROI analysis)	D8

Three deliberate symbol reuses deserve a caveat, recorded here so that no derivation needs to repeat it. First, M names both the loop Jacobian and the exploration second moment; the design M occurs only in D4/D5, and context disambiguates the two. Second, κ is the correction sensitivity $|dh_{\text{PO}}/d\varepsilon|$ of D8/D9, not a multi-dealer spillover parameter: no such symbol occurs in this paper. Third, ϕ is the noise-feedback gain of D2, while the T9 material in D3 separately writes ϕ_u and ϕ_0 for elements of the sensitivity span, a self-contained reuse local to that subsection.

Assumption register. Assumptions are cited by number (A1 through A6) throughout the record.

- **A1 (local quadratic scope).** Φ is C^3 and strongly concave on the operating interval; all expansions are taken at h^* , with the third-order remainder controlled through $L_3 = \sup|\Phi'''|$. The identities of the record are exact for the quadratic model and first-order accurate otherwise. As a matter of method, the nonlinear deviation is measured (the drift study), never assumed away.
- **A2 (exploration classes).** All policies are non-anticipating: u_t is $\mathcal{F}_t$ -measurable. Two subclasses are used. *A2-sym*, conditionally symmetric exploration, $\mathbb{E}[u_t | \mathcal{F}_t] = 0$, under which the linear term of L1 vanishes in expectation. *A2-adaptive*, unrestricted non-anticipating exploration, under which costs are measured from the certainty-equivalent anchor (D0 Sec. 4; convention 2 below) and the exploitation lemma L2 of D3b applies.
- **A3 (noise exogeneity, with its named relaxation).** The response noise ζ_t is iid, mean zero, with variance σ^2 , independent of $\mathcal{F}_t$. Relaxation A3': the feedback channel $\phi \zeta_t$ enters d_{t+1} . Under A3' the exact $O(1/T)$ bias of D2 Sec. 4 applies, with its $(3m - 1)$ sign structure (P2.2), and “exploration-dominant design” means $\sigma_e^2 \gg \phi^2 \sigma^2$.
- **A4 (trust region).** $|d_t| \leq r$ pathwise. The radius r enters the constants of D3b (L2, T4), the feasible set of the design problems (T5), and the basin argument of D6.
- **A5 (stationary-scale exploration; D3b only).** $S_T = \Theta(T)$, with $S_T = \sum_{t \leq T} d_t^2$ the cumulative design energy. The transient-dominated regime $S_T = o(T)$ is governed by D1 (T1) instead.
- **A6 (stable base loop).** $m < 1$ in the scalar case and $\rho(M) < 1$ in the d -dimensional case, for all stationary-regime statements. Boundary behaviour ($m = 1$) appears only in the degeneracy statement C1.2.

The two anchoring conventions (audit-critical). C_T is an incremental cost, and its value depends on the baseline against which the increment is measured. The record fixes one anchor per exploration class; these are the two audit-critical definitions.

Convention 1 (incremental anchor; all A2-sym results). C_T is measured relative to the jitter-free loop at h^* , that is, the same retraining loop run without exploration. Anchoring at h_{PO} instead adds the sunk echo-chamber term and provably destroys the invariance of D2 (T2): with anchor gap $g = h^* - h_{\text{PO}}$, the anchored exchange-rate product is inflated by exactly the factor

$$1 + \frac{g^2}{v},$$

663 a computation verified in D0. The formal proof is recorded in the proofs appendix.

664 *Convention 2 (certainty-equivalent anchor; all A2-adaptive results).* C_T is measured relative to the
 665 deployment that is optimal under current posterior knowledge. Exploitation of learned structure is
 666 thereby re-anchored away, and D3b (L2, T4) prices only the unlearned residual.

667 The two anchors coincide under A2-sym with an uninformative start, so the conventions agree on the
 668 class where both are defined.

669 **D0: The model, the two budgets, and the cost-equivalence lemma (L1)**

670 This subsection records the local model, the two exploration budgets, and the computation behind the
 671 cost-equivalence lemma L1. It is derived in full and verified numerically (`verify_all.py`, section
 672 V0). Everything downstream (D1 through D9) consumes the definitions and the lemma established
 673 here. Notation follows the REFLEX market model [Nagarajan and Ashok, 2026] ($\gamma, \beta, \varepsilon, \psi, h_{SP},$
 674 h_{PO}).

675 **The local model.** Work in a neighbourhood of the performatively stable point h^* , the RRM fixed
 676 point of that market [Nagarajan and Ashok, 2026], and write deviations $d_t := h_t - h^*$. The linearized
 677 closed loop is

$$\begin{aligned} d_{t+1} &= -m d_t + u_t && \text{(retraining cobweb plus exploration),} \\ \tau_t &= \tau(h^*) - \varepsilon d_t + \zeta_t && \text{(observed flow).} \end{aligned}$$

678 Here $m = \varepsilon\beta/\gamma \in [0, 1)$ is the retraining modulus, whose closed form comes with the model
 679 [Nagarajan and Ashok, 2026]. The exploration input u_t is chosen by the operator: it may
 680 be iid noise ($u_t = \sigma_e \xi_t$), a deterministic schedule, or adaptive, that is $\mathcal{F}_t$ -measurable with
 681 $\mathcal{F}_t = \sigma(d_0, u_0, \dots, u_{t-1}, \tau_0, \dots, \tau_{t-1})$. The response noise ζ_t is iid, mean zero, with variance
 682 σ^2 , independent of $\mathcal{F}_t$; the feedback violation of this independence assumption is quantified in the
 683 exchange-rate derivation D2 (the feedback-bias result P2.2). The estimand is the response slope ε ,
 684 scalar here; the d -dimensional version (D4, behind T5) and the structural-family version (D5, behind
 685 T6) are treated in their own derivation subsections.

686 The true performative objective $\Phi(h) = J(h; \tau(h))$ (REFLEX theory 1.2) is smooth and strongly
 687 concave on the operating interval, with

$$\Phi'(h^*) = \Delta(h^*) = -\beta(h^* - \psi)\varepsilon(h^*) =: \delta_0, \quad \Phi''(h^*) = -\gamma_{PO}$$

688 (a direct computation in the market model of Nagarajan and Ashok [2026]). The slope δ_0 is nonzero
 689 precisely because $h^* \neq h_{PO}$.

690 **The two budgets.** Two notions of exploration budget are used throughout:

$$\begin{aligned} D_T &:= \sum_{t \leq T} d_t^2 && \text{(deviation budget: realized design energy),} \\ C_T &:= \mathbb{E} \left[\sum_{t \leq T} (\Phi(h^*) - \Phi(h_t)) \right] && \text{(value budget: expected incremental cost).} \end{aligned}$$

691 D_T is the tracking-error form, a desk's deviation limit: it is observable, its meaning is policy-
 692 independent, and it is the correct constraint for the exact minimax theorem T3 (derivation D3). C_T is
 693 the realized-P&L form. Lemma L1 relates the two.

694 **Lemma L1 (cost equivalence).** For any exploration rule that is non-anticipating and conditionally
 695 symmetric, meaning $\mathbb{E}[u_t | \mathcal{F}_t] = 0$ (for example iid exploration, or a scheduled sign-symmetric
 696 jitter), in the regime where third-order terms are negligible,

$$C_T = \frac{1}{2} \gamma_{PO} \mathbb{E}[D_T] + R_T, \quad |R_T| \leq \frac{L_3}{6} \mathbb{E} \left[\sum_{t \leq T} |d_t|^3 \right],$$

697 with $L_3 = \sup |\Phi'''|$ on the operating interval. The linear term of the expansion contributes exactly
 698 zero to the expectation, and contributes a fluctuation of standard deviation $|\delta_0| \sqrt{\text{Var}(\sum_{t \leq T} d_t)}$ to
 699 the realized cost.

700 **Proof.** Taylor-expand Φ at h^* :

$$\Phi(h^*) - \Phi(h_t) = -\delta_0 d_t + \frac{1}{2} \gamma_{\text{PO}} d_t^2 + r_t, \quad |r_t| \leq \frac{L_3}{6} |d_t|^3,$$

701 then sum over $t \leq T$ and take expectations. For the linear term: d_t is built from $\{u_s\}_{s < t}$ through
 702 the linear recursion, so the mean obeys $\mathbb{E}[d_{t+1}] = -m \mathbb{E}[d_t] + \mathbb{E}[u_t]$; conditional symmetry gives
 703 $\mathbb{E}[u_t] = \mathbb{E}[\mathbb{E}[u_t | \mathcal{F}_t]] = 0$, hence $\mathbb{E}[d_t] = (-m)^t \mathbb{E}[d_0]$. Starting at the operating point ($d_0 = 0$),
 704 or in the stationary regime, $\mathbb{E}[d_t] = 0$ for all t , so $\mathbb{E}[\sum_{t \leq T} \delta_0 d_t] = 0$. The quadratic term yields
 705 $\frac{1}{2} \gamma_{\text{PO}} \mathbb{E}[D_T]$, and the remainder is bounded by the stated third-order term. $\square$

706 **Fluctuation decomposition (exact, and a corrected earlier claim).** The realized cost's variance
 707 splits into two terms with zero cross term: the cross-moment sums built from $\mathbb{E}[d_t d_s^2]$ vanish by the
 708 sign symmetry of the stationary law. Exactly,

$$\text{Var}(C_T^{\text{real}}) = \delta_0^2 \text{Var}\left(\sum_{t \leq T} d_t\right) + \frac{\gamma_{\text{PO}}^2}{4} \text{Var}\left(\sum_{t \leq T} d_t^2\right),$$

709 with the long-run scales, v denoting the stationary variance of d_t ,

$$\begin{aligned} \text{Var}\left(\sum_{t \leq T} d_t\right) &\sim T v \frac{1-m}{1+m} && \text{(long-run variance: negative autocorrelation helps),} \\ \text{Var}\left(\sum_{t \leq T} d_t^2\right) &\sim T \frac{2v^2(1+m^2)}{1-m^2} && \text{(Gaussian stationary AR(1)).} \end{aligned}$$

710 A caveat, recorded per the falsification convention: an earlier draft asserted that the linear term always
 711 dominates. The first verification run falsified that at moderate δ_0 (measured sd ratio 1.62, exactly
 712 reproduced by the two-term formula above), and the claim is now stated as the full decomposition.
 713 Which term dominates is decided by comparing $\delta_0^2(1-m)/(1+m)$ against $(\gamma_{\text{PO}}^2 v/2)(1+m^2)/(1-m^2)$,
 714 both computable. V0 now verifies the decomposition itself.

715 **Why the anchoring is load-bearing (recorded from the audit).** Anchoring the cost at the perfor-
 716 mative optimum h_{PO} instead of h^* adds terms of the type $T(h^* - h_{\text{PO}})^2$ that are paid whether or not
 717 the operator explores: the sunk echo-chamber cost of blindness (REFLEX theory 1.2, 6b). Numeri-
 718 cally (verify_m2_identity.py, Claim 3): writing $g = h^* - h_{\text{PO}}$ for the gap, the mis-anchored
 719 product equals

$$\frac{1}{2} \gamma_{\text{PO}} \sigma^2 \left(1 + \frac{g^2}{v}\right),$$

720 which destroys the exchange-rate invariance of D2 (theorem T2). The incremental anchor at h^* is the
 721 definition under which every later theorem is true.

722 **The certainty-equivalent anchor (needed for the adaptive minimax bound T4, derivation D3).**
 723 For adaptive policies the symmetric-exploration condition can be violated deliberately: since $\delta_0 \neq 0$,
 724 drifting toward h_{PO} earns first-order value. Two observations discipline this. First, decompose
 725 $\Phi'(h^*)$ as a function of the estimand:

$$\varphi(\varepsilon) := \Phi'(h^*; \varepsilon) = -\beta(h^* - \psi)\varepsilon,$$

726 which is linear in ε . Under a prior centred at $\bar{\varepsilon}$, split $\varphi(\varepsilon) = \varphi(\bar{\varepsilon}) + \varphi'(\bar{\varepsilon})(\varepsilon - \bar{\varepsilon})$. Second, the
 727 first (known) part is exploitable without any learning: it is a pure consequence of already-known
 728 information, that is, of choosing the anchor wrongly. Define therefore the certainty-equivalent anchor
 729 h_{CE}^* : the deployment that is optimal under current knowledge (the posterior mean), with exploration
 730 cost measured relative to the trajectory that sits at h_{CE}^* . Relative to that anchor the residual linear
 731 coefficient is

$$\varphi'(\bar{\varepsilon})(\varepsilon - \bar{\varepsilon}) = -\beta(h^* - \psi)(\varepsilon - \bar{\varepsilon}),$$

732 which has zero prior mean: only genuinely unknown structure can be exploited, and the exploitation-
 733 information lemma L2 (in D3, feeding T4) bounds that exploitation by the information already
 734 purchased. This definition also matches practice: a desk's exploration cost is measured against its
 735 own current best policy, not against an oracle's.

736 **Numerical verification (V0).** Three checks close the record for this subsection.

- 737 • $\mathbb{E}[C_T] = \frac{1}{2} \gamma_{\text{PO}} \mathbb{E}[D_T]$ holds within Monte Carlo error, for a quadratic-plus-linear objective
738 with $\delta_0 \neq 0$, under iid exploration.
- 739 • The realized-cost fluctuation scale matches $|\delta_0| \text{sd}(\sum_{t \leq T} d_t)$: the linear term is a variance
740 contribution, not a bias.
- 741 • A deliberately asymmetric (drifting) exploration rule breaks the lemma in the predicted di-
742 rection and by the predicted first-order amount $-\delta_0 \mathbb{E}[\sum_{t \leq T} d_t]$: the conditional-symmetry
743 condition of L1 is necessary, not decorative.

744 **D1: Information saturation by the Lyapunov identity (T1, C1.1, C1.2, R1)**

745 This subsection records the derivation behind T1 and its corollaries C1.1 and C1.2, together with
746 the non-normal refinement R1. Status of the record: the scalar closed form is verified to 10^{-16} , and
747 the d -dimensional identity, including the non-normal amplification, is verified numerically (V1); the
748 proofs are complete.

749 **Scalar warm-up (the verified base case).** With no exploration ($u_t = 0$) and modulus $m < 1$, the
750 deviation obeys $d_t = (-m)^t d_0$, so the design energy accumulated by horizon T is

$$D_T = \sum_{t=0}^T d_t^2 = d_0^2 \frac{1 - m^{2(T+1)}}{1 - m^2} \rightarrow \frac{d_0^2}{1 - m^2} \quad (T \rightarrow \infty).$$

751 The Fisher information for the response slope ε from OLS on the trajectory is $S_{xx}/\sigma^2 \leq D_\infty/\sigma^2$:
752 bounded uniformly in the horizon. Hence, by Cramér–Rao, for any unbiased estimator and at every
753 horizon,

$$\text{Var}(\hat{\varepsilon}) \geq \frac{\sigma^2 (1 - m^2)}{d_0^2}.$$

754 This is the scalar case of T1; with $d_0 = h_0 - h^*$ the cap $d_0^2/(1 - m^2)$ is the quantity in the register
755 statement of C1.1.

756 **C1.1 (safety implies blindness).** The cap $d_0^2/(1 - m^2)$ is strictly increasing in m : the fast-
757 contracting (safe) loop reveals least about its own performativity, and identifiability is a near-instability
758 phenomenon. This is immediate from the closed form above; the economic content is stated in the
759 paper body.

760 **C1.2 (design degeneracy).** At $m = 1$ the noiseless iterates form the period-two orbit $\{+d_0, -d_0\}$;
761 for $m < 1$ the support collapses geometrically onto $\{0\}$. In either case the empirical design measure
762 has at most two effective support points, which by the Chebyshev counting argument of D5 (theorem
763 T6) cannot identify a three-parameter response family. Curvature is invisible to retraining trajectories
764 at every m .

765 **The d -dimensional identity via the discrete Lyapunov equation.** Let the linearized joint loop be
766 $d_{t+1} = M d_t$ with $M \in \mathbb{R}^{d \times d}$ and $\rho(M) < 1$; in the multi-bond and multi-dealer instantiations M
767 comes from the REFLEX market model [Nagarajan and Ashok, 2026] and is generally not symmetric.

768 **Theorem (exact energy; T1).** The total design energy is the quadratic form

$$E(d_0) := \sum_{t \geq 0} \|M^t d_0\|^2 = d_0^\top P d_0,$$

769 where P is the unique positive semidefinite solution of the discrete Lyapunov equation

$$P = I + M^\top P M.$$

770 **Proof.** Set $P := \sum_{t \geq 0} (M^\top)^t M^t$; the series converges absolutely if and only if $\rho(M) < 1$ (Gelfand’s
771 spectral radius formula). Then

$$M^\top P M = \sum_{t \geq 0} (M^\top)^{t+1} M^{t+1} = P - I,$$

772 so P solves the equation. Uniqueness: the difference Q of two solutions satisfies $Q = M^\top Q M =$
 773 $(M^\top)^t Q M^t \rightarrow 0$. Finally, $d_0^\top P d_0 = \sum_{t \geq 0} \|M^t d_0\|^2$ by expanding the sum term by term. $\square$

774 Scalar check: $P = 1/(1 - m^2)$, recovering the warm-up above.

775 **Directional information.** For the response slope in a direction u (the observation $\tau_t = \tau^* -$
 776 $\varepsilon u^\top d_t + \zeta_t$ probes the component along u), the relevant energy is

$$\sum_{t \geq 0} (u^\top M^t d_0)^2 = d_0^\top P_u d_0, \quad P_u = uu^\top + M^\top P_u M,$$

777 the same Lyapunov structure with a rank-one right-hand side. By Cramér–Rao this yields $\text{Var}(\hat{\varepsilon}) \geq$
 778 $\sigma^2 (d_0^\top P_u d_0)^{-1}$ for the component along any direction u , which is the bound stated in T1. Hence the
 779 Fisher information is bounded in every direction, and the information operator satisfies $\mathcal{I} \preceq P_{\text{full}}/\sigma^2$,
 780 where P_{full} solves the matrix-valued Lyapunov equation. Saturation is not a scalar accident.

781 **Non-normality: transiently amplified information (R1).** If M is normal, P has eigenvalues
 782 $1/(1 - \lambda_i^2)$ and the energy is governed by the spectrum alone. If M is non-normal (heterogeneous
 783 multi-dealer Jacobians are), transient growth inflates the energy beyond the spectral prediction:

$$E(d_0) \leq \kappa(V)^2 \frac{\|d_0\|^2}{1 - \rho(M)^2},$$

784 where V is the eigenbasis of M and $\kappa(V)$ its condition number, and the inequality is achieved in
 785 order: for the Jordan-type family

$$M = \begin{pmatrix} m & a \\ 0 & m \end{pmatrix},$$

786 the energy along the first coordinate grows like $a^2/(1 - m^2)^3$ for large a , verified numerically in V1
 787 against the exact Lyapunov solve. Badly-conditioned ecosystems thus leak extra information about
 788 themselves during transients, the free-information remark of the paper, now with an exact constant
 789 (P itself). The paper states the exact P -form and keeps the pseudospectral refinement for the journal
 790 version.

791 **Scope: what T1 does not say.** Two caveats are part of the record and are stated in the paper.

- 792 • With exploration on, saturation is replaced by the exchange rate of D2 (T2): the transient
 793 information $d_0^\top P d_0/\sigma^2$ is the free part; everything after is bought.
- 794 • The bound is for the linearized loop; the simulator’s nonlinear regimes are handled empiri-
 795 cally (the drift study, under the D2/V2 protocol).

796 **Numerical verification (V1).** Four checks anchor the derivation. (i) The scalar closed form against
 797 the direct sum: exact to floating point. (ii) Random stable M of sizes 2×2 and 5×5 : the direct
 798 energy sum against the Lyapunov solve (a `scipy`-free fixed-point iteration), agreement to 10^{-12} . (iii)
 799 Non-normal amplification: the energy for the family $M = \begin{pmatrix} m & a \\ 0 & m \end{pmatrix}$ grows as a^2 , far exceeding the
 800 normal-matrix bound $1/(1 - m^2)$, and matches $d_0^\top P d_0$ exactly. (iv) The directional version: the
 801 rank-one Lyapunov solution P_u against the direct directional sum.

802 D2: The exchange-rate identity and its concentration (T2, P2.1, P2.2)

803 This subsection records the derivations behind T2 (the exchange-rate identity in pathwise form), P2.1
 804 (its concentration), and P2.2 (the feedback-bias constant). Status of record: the identity was verified
 805 to zero relative error by a pre-existing check; the pathwise refinement, the concentration rate, and the
 806 feedback-bias constant are derived here and verified in V2. The formal statements live in the register
 807 of theorem statements, and the proofs in the proofs appendix.

808 **Setup.** The stationary jittered loop is

$$d_{t+1} = -m d_t + \sigma_e \xi_t, \quad v = \text{Var}(d_t) = \frac{\sigma_e^2}{1 - m^2}.$$

809 OLS of the response τ_t on the quote h_t has conditional-on-design variance $\text{Var}(\hat{\varepsilon} \mid \text{design}) = \sigma^2/S_{xx}$
 810 with $S_{xx} = \sum_{t \leq T} (d_t - \bar{d})^2$, and the cost of Lemma L1 is $\frac{1}{2} \gamma_{\text{PO}} \sum_{t \leq T} d_t^2$.

811 **The identity, upgraded from expectation to pathwise (T2). Proposition (pathwise identity).**
812 On every realization,

$$\begin{aligned} \text{Var}(\hat{\varepsilon} \mid \text{design}) \times \frac{1}{2} \gamma_{\text{PO}} \sum_{t \leq T} d_t^2 &= \frac{1}{2} \gamma_{\text{PO}} \sigma^2 \frac{\sum_{t \leq T} d_t^2}{S_{xx}} \\ &= \frac{1}{2} \gamma_{\text{PO}} \sigma^2 \left(1 + \frac{T \bar{d}^2}{S_{xx}} \right). \end{aligned}$$

813 Since $\bar{d} \rightarrow 0$ almost surely at rate $T^{-1/2}$ in the stationary regime, the product equals $\frac{1}{2} \gamma_{\text{PO}} \sigma^2 (1 +$
814 $O_P(1/T))$ pathwise: a strictly stronger statement than the in-expectation identity, and the form the
815 experiments actually measure. The correction term $T \bar{d}^2 / S_{xx}$ is nonnegative, so the realized product
816 can only sit above the constant, by a vanishing amount.

817 **Invariance.** Neither σ_e , nor m , nor T appears: more jitter buys information and burns value at the
818 same rate; a larger modulus amplifies the jitter into more excitation and more cost at the same rate.
819 The constant $\frac{1}{2} \gamma_{\text{PO}} \sigma^2$ is the exchange rate between self-knowledge and value.

820 **Concentration: the identity is observable (P2.1).** The realized value cost (Lemma L1) carries
821 the linear-term fluctuation $-\delta_0 \sum_{t \leq T} d_t$. In the stationary regime $\sum_t d_t$ is a mean-zero weakly
822 dependent sum, so

$$\frac{\text{sd}(\text{realized } C_T)}{\mathbb{E}[C_T]} = O(T^{-1/2}),$$

823 with the explicit leading constant given by the AR(1) long-run variance $v(1 + \rho)/(1 - \rho)$ at lag-one
824 autocorrelation $\rho = -m$, that is $v(1 - m)/(1 + m)$:

$$\text{Var} \left(\sum_{t \leq T} d_t \right) = T v \frac{1 - m}{1 + m} (1 + o(1)).$$

825 The negative autocorrelation of the cobweb shrinks the fluctuation below the iid level $T v$: a small
826 bonus worth one sentence in the paper. Hence the measured product concentrates around the invariant
827 at rate $T^{-1/2}$. Verified in V2: the seed-variance of the measured product scales as $1/T$ (it halves
828 when T doubles, a factor-4 drop when T quadruples, within Monte-Carlo tolerance).

829 **Detail: S_{xx} versus $\sum_t d_t^2$.** The estimator uses the centered S_{xx} ; the cost uses the raw $\sum_t d_t^2$. Their
830 ratio is

$$\frac{\sum_{t \leq T} d_t^2}{S_{xx}} = 1 + \frac{T \bar{d}^2}{S_{xx}} = 1 + O_P(1/T),$$

831 as above. Statements in the paper use whichever form is natural and record this $O(1/T)$ equivalence
832 once, in the present form.

833 **The feedback (predetermined-regressor) bias, with its constant (P2.2).** In the real loop the
834 observed response feeds the next deployment. Model this as

$$d_{t+1} = -m d_t + \sigma_e \xi_t + \phi \zeta_t,$$

835 with ϕ the noise-feedback gain: retraining on the realized flow moves the next quote by $\phi \zeta_t$. Now d_t
836 is correlated with past ζ , so the regressor is predetermined but not strictly exogenous.

837 **Proposition (Stambaugh-type bias, full form, corrected by verification).** The OLS slope satisfies

$$\mathbb{E}[\hat{\varepsilon} - (-\varepsilon)] = \frac{\phi \sigma^2 (3m - 1)}{(1 - m^2) T v_\phi} (1 + o(1)), \quad v_\phi = \frac{\sigma_e^2 + \phi^2 \sigma^2}{1 - m^2},$$

838 where $-\varepsilon$ is the true slope in this sign convention (written $\varepsilon_{\text{true}}$ in P2.2) and v_ϕ is the stationary
839 variance of the feedback loop. The bias is $O(1/T)$, proportional to the feedback gain, and changes
840 sign at $m = 1/3$: slow-contracting loops bias the slope estimate upward, fast-contracting ones
841 downward, and at $m = 1/3$ exactly the feedback bias vanishes at first order.

842 **Derivation, both terms.** (Honesty record, kept per the falsification convention: the first draft retained
843 only the centering term and was falsified by V2; the covariance term below is the correction.) Write
844 $\hat{b} - b = N/S$ with $N = \sum_{t \leq T} (d_t - \bar{d}) \zeta_t$ and $S = S_{xx}$, and expand

$$\mathbb{E}\left[\frac{N}{S}\right] = \frac{\mathbb{E}[N]}{\mathbb{E}[S]} - \frac{\text{Cov}(N, S)}{\mathbb{E}[S]^2} + O(T^{-2}).$$

845 *Centering term.* Here $\mathbb{E}[d_t \zeta_t] = 0$, since ζ_t is independent of the past-built d_t , so $\mathbb{E}[N] =$
846 $-\mathbb{E}[\bar{d} \sum_t \zeta_t]$. With $\mathbb{E}[d_s \zeta_t] = \phi \sigma^2 (-m)^{s-t-1}$ for $s > t$,

$$\mathbb{E}[N] = -\frac{1}{T} \sum_{s>t} \phi \sigma^2 (-m)^{s-t-1} = -\frac{\phi \sigma^2}{1+m} + O(1/T).$$

847 *Covariance term (the one the first draft missed).* By the Gaussian fourth-moment (Wick) expansion,

$$\text{Cov}(N, S) = 2 \sum_{t,s} \mathbb{E}[\tilde{d}_t \tilde{d}_s] \mathbb{E}[\tilde{d}_s \zeta_t], \quad \tilde{d} = d - \bar{d};$$

848 substituting $\mathbb{E}[\tilde{d}_t \tilde{d}_s] \simeq v_\phi (-m)^{|s-t|}$ and the cross-covariance above,

$$\text{Cov}(N, S) = 2 v_\phi \phi \sigma^2 T \sum_{k \geq 1} (-m)^k (-m)^{k-1} = -\frac{2 v_\phi \phi \sigma^2 T m}{1-m^2}.$$

849 *Combine.* With $\mathbb{E}[S] \simeq T v_\phi$,

$$\text{bias} = \left[-\frac{1}{1+m} + \frac{2m}{1-m^2} \right] \frac{\phi \sigma^2}{T v_\phi} = \frac{3m-1}{1-m^2} \cdot \frac{\phi \sigma^2}{T v_\phi},$$

850 which is P2.2. Verified in V2 at two moduli straddling the sign change: $m = 0.5$ gives a positive bias
851 and $m = 0.2$ a negative one, each within Monte-Carlo tolerance of the constant. The sign change is
852 the sharpest possible falsification test of the two-term structure, and it passes.

853 **Paper usage.** State the identity (T2) under exploration-dominant design; report the bias formula
854 (P2.2) as the known, computable deviation, a checkable condition (ϕ is estimable from the loop's
855 own update rule), not an assumption.

856 **Verified numerically (V2).** The V2 checks for this subsection: (1) the pathwise product equals
857 $\frac{1}{2} \gamma_{\text{PO}} \sigma^2 (1 + T \bar{d}^2 / S_{xx})$ exactly; (2) product concentration: the seed-variance of the measured
858 product scales as $1/T$, a factor-4 drop when T quadruples, within Monte-Carlo tolerance; (3) the
859 long-run variance of $\sum_t d_t$ matches $T v (1-m)/(1+m)$; (4) the feedback bias: sign, magnitude,
860 and $1/T$ scaling of $\phi \sigma^2 (3m-1)/((1-m^2) T v_\phi)$.

861 **D3: Minimax lower bounds and the exact reach (T3, L2, T4, T9)**

862 This record derives the two lower bounds that turn the exchange-rate identity of D2 (register T2) from
863 a property of one scheme into a property of the problem, and then the exact reach within which that
864 floor is structure-proof against a policy that knows the response family (T9). T2 shows that a specific
865 scheme, stationary symmetric jitter with ordinary least squares, achieves $\text{Var}(\hat{\varepsilon}) \times \text{cost} = \frac{1}{2} \gamma_{\text{PO}} \sigma^2$.
866 The claim that makes this an exchange rate is that no adaptive exploration policy and no estimator
867 does better. There are two versions of the bound, one per budget type: an exact deviation-budget
868 bound (T3, derived in full and verified numerically in V3a) and a value-budget bound (T4), whose
869 engine is the exploitation-information lemma (L2), proved below with its $O(T^{-1/2})$ correction
870 established and verified in V3b. Throughout, the formal proofs are recorded in the proofs appendix;
871 this subsection records the derivations and the caveats attached to them.

872 **Deviation budget: van Trees over adaptive designs (T3).** Setting: the exploration policy is
873 arbitrary and adaptive (each u_t any measurable function of the history), subject to the pathwise budget
874 $\sum_{t \leq T} d_t^2 \leq D$ almost surely; the estimator $\hat{\varepsilon}$ of ε is arbitrary, bias allowed. The claim (T3) is that
875 the minimax risk over an interval of ε values of width w satisfies

$$\inf_{\text{policy, estimator}} \sup_{\varepsilon} \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq \frac{\sigma^2}{D + \sigma^2(\pi/w)^2} = \frac{\sigma^2}{D} (1 - o(1)) \quad \text{as } D(w/\sigma)^2 \rightarrow \infty.$$

876 Multiplying by the budget cost $\frac{1}{2}\gamma_{\text{PO}}D$ (cost equivalence, L1) gives

$$\sup_{\varepsilon} \text{Var}(\hat{\varepsilon}) \times \frac{1}{2}\gamma_{\text{PO}}D \geq \frac{1}{2}\gamma_{\text{PO}}\sigma^2(1 - o(1)) :$$

877 the exchange rate is a floor.

878 The derivation is the van Trees inequality (Bayesian Cramér–Rao, in the form of Gill and Levit, 1995)
879 with a prior density π_0 supported on the width- w interval:

$$\mathbb{E}_{\pi_0} \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq \frac{1}{\mathbb{E}_{\pi_0}[I_T(\varepsilon)] + I(\pi_0)}, \quad I(\pi_0) = \int \frac{(\pi'_0)^2}{\pi_0},$$

880 where $I(\pi_0)$ is the prior information, equal to $(\pi/w)^2$ for the cosine-squared prior (the minimizer of
881 prior information on an interval of width w), and $I_T(\varepsilon)$ is the Fisher information of the adaptively
882 designed experiment. The one nonstandard step is that I_T for an adaptive design is still bounded by
883 the design energy: by the chain rule for Fisher information over the filtration,

$$I_T(\varepsilon) = \sum_{t \leq T} \mathbb{E}[I(\tau_t | \mathcal{F}_t; \varepsilon)] = \sum_{t \leq T} \frac{\mathbb{E}[d_t^2]}{\sigma^2},$$

884 because conditionally on $\mathcal{F}_t$ the observation τ_t is Gaussian with mean linear in ε (slope $-d_t$, with d_t
885 $\mathcal{F}_t$ -measurable): the policy can place information but cannot manufacture it. The pathwise budget
886 then gives $\mathbb{E}[I_T] \leq D/\sigma^2$; substituting, and taking the supremum over the interval ($\sup \geq \mathbb{E}_{\pi_0}$),
887 yields the bound.

888 Three remarks recorded with the derivation. (i) Biased estimators are covered: van Trees needs no
889 unbiasedness. (ii) The bound is achieved, up to the $o(1)$, by two-point symmetric probing at the
890 budget with least squares, tying back to the D2 identity (T2); this is the “attained with equality”
891 clause of T3. (iii) The $o(1)$ is explicit: it equals $\sigma^2(\pi/w)^2/D$.

892 **D3b, the adaptive companion. Value budget: the exploitation problem.** Under the value budget
893 the policy pays $\Phi(h_{\text{CE}}^*) - \Phi(h_t)$ per step (the certainty-equivalent anchor of D0). The residual linear
894 coefficient at the anchor is

$$\varphi'(\bar{\varepsilon})(\varepsilon - \bar{\varepsilon}) = -\beta(h^* - \psi)(\varepsilon - \bar{\varepsilon}),$$

895 so a policy could try to earn while exploring: guess the sign of $\varepsilon - \bar{\varepsilon}$ and drift accordingly. The
896 lemma below says this self-financing is bounded by the information already purchased: the policy
897 cannot exploit what it has not yet paid to know.

898 Setting: symmetric two-point prior $\varepsilon \in \{\bar{\varepsilon} - \delta, \bar{\varepsilon} + \delta\}$; write $s = \text{sign}(\varepsilon - \bar{\varepsilon})$ and $g_1 := \beta|h^* - \psi|$,
899 the known coefficient of the unknown part. The policy is non-anticipating, with per-step trust region
900 $|d_t| \leq r$; write $S_t := \sum_{i \leq t} d_i^2$ for the design energy.

901 **The exploitation–information lemma (L2).** Two ingredients. First, the posterior-imbalance step:
902 the policy’s knowledge of s at time t obeys

$$|\mathbb{E}[s | \mathcal{F}_t]| \leq \text{TV}(P_t^+, P_t^-) =: \text{TV}_t$$

903 (posterior imbalance is bounded by the distinguishability of the two environments; the starting prior
904 is symmetric). Register L2 records the averaged form of this step as the exact identity $\mathbb{E}|\mathbb{E}[s | \mathcal{F}_t]| =$
905 $\text{TV}(P_t^+, P_t^-)$. Second, the Pinsker step: the two environments differ only in the observation means,
906 by $2\delta d_i$ at step i , so

$$\text{KL}_t = \sum_{i \leq t} \frac{(2\delta d_i)^2}{2\sigma^2} = \frac{2\delta^2 S_t}{\sigma^2}, \quad \text{TV}_t \leq \sqrt{\text{KL}_t/2} = \frac{\delta\sqrt{S_t}}{\sigma}.$$

907 (For adaptive designs the same computation runs through the chain rule for KL over the filtration,
908 with S_t replaced by $\mathbb{E}S_t$, exactly as in the Fisher computation for T3; this is the form registered in
909 L2.)

910 Chaining these, the expected exploitation gain obeys, for any non-anticipating policy,

$$\begin{aligned}\mathbb{E}[\text{gain}_T] &:= \mathbb{E}\left[\sum_{t \leq T} g_1 \delta s d_t\right] \leq g_1 \delta \sum_{t \leq T} \mathbb{E}[\text{TV}_t | d_t|] \\ &\leq \frac{g_1 \delta^2}{\sigma} \sqrt{S_T} \sum_{t \leq T} \mathbb{E}|d_t| \leq \frac{g_1 \delta^2}{\sigma} S_T \sqrt{T},\end{aligned}$$

911 where the first inequality is the tower rule plus the posterior-imbalance step, the second uses $\text{TV}_t \leq$
 912 $\delta \sqrt{S_t}/\sigma \leq \delta \sqrt{S_T}/\sigma$, and the last is Cauchy–Schwarz ($\sum_{t \leq T} |d_t| \leq \sqrt{T} S_T$). For fully adaptive
 913 designs the same chain, run with expected energies, gives the closed form recorded in L2, $\mathbb{E}[\text{gain}_T] \leq$
 914 $g_1 \delta^{3/2} T^{1/2} (\mathbb{E} S_T)^{3/4} / \sigma^{1/2}$; both forms are $O(\sigma \sqrt{T})$ at the minimax scale below. The formal proof
 915 is recorded in the proofs appendix under L2.

916 **The minimax scale and the value-budget bound (T4).** The minimax-relevant prior scale is
 917 $\delta = c \sigma / \sqrt{S_T}$: this is the accuracy the budget can buy (any larger δ is learned exactly, any smaller is
 918 unlearnable). Substituting into the lemma,

$$\mathbb{E}[\text{gain}_T] \leq g_1 c^2 \sigma \sqrt{T}, \quad \text{while} \quad \mathbb{E}[C_T] = \frac{1}{2} \gamma_{\text{PO}} S_T \sim \frac{1}{2} \gamma_{\text{PO}} T v$$

919 (cost equivalence L1, stationary-scale exploration $S_T = \Theta(T)$ with per-step exploration variance v).
 920 The exploitable fraction of the budget is therefore

$$\frac{\mathbb{E}[\text{gain}_T]}{\mathbb{E}[C_T]} \leq \frac{2 g_1 c^2 \sigma}{\gamma_{\text{PO}} v \sqrt{T}} = O(T^{-1/2}).$$

921 The Le Cam two-point reduction then quantifies the constant. At the minimax scale $\text{TV}_T \leq c$, so any
 922 estimator’s worst-case risk over the pair obeys

$$\sup_{\varepsilon} \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \geq \frac{\delta^2}{2} (1 - \text{TV}_T) \geq \frac{\delta^2}{2} (1 - c).$$

923 Multiplying by $\mathbb{E}[C_T] = \frac{1}{2} \gamma_{\text{PO}} S_T$, using $\delta^2 S_T = c^2 \sigma^2$, and applying the exploitation rebate (the
 924 value budget can be underpaid only by the exploited gain, an $O(T^{-1/2})$ fraction by the display
 925 above):

$$\sup \text{Var}(\hat{\varepsilon}) \times C_T \geq \frac{\gamma_{\text{PO}} \sigma^2}{4} c^2 (1 - c) (1 - O(T^{-1/2})).$$

926 The factor $c^2(1 - c)$ is maximized at $c = 2/3$, with value $4/27$, giving the registered bound of T4:

$$\inf_{\text{policy, estimator}} \sup \text{Var}(\hat{\varepsilon}) \times C_T \geq \frac{\gamma_{\text{PO}} \sigma^2}{27} (1 - O(T^{-1/2})).$$

927 The sharp constant $\frac{1}{27}$, matching the achievable product of T2, that is $\inf \sup \text{Var}(\hat{\varepsilon}) \times C_T \geq$
 928 $\frac{1}{2} \gamma_{\text{PO}} \sigma^2 (1 - O(T^{-1/2}))$, is conjectured and is part of OPEN-1: no simulated policy fell below it by
 929 more than Monte-Carlo error (run_open1.py, part B).

930 **Interpretation: the self-referential structure.** The profitable direction is s , the very thing being
 931 estimated. Pinsker converts “how much the policy knows about s ” into “how much design energy it
 932 has already spent”; any gain is therefore a rebate proportional to information already paid for, and the
 933 rebate’s rate vanishes. Knowledge cannot be bootstrapped from its own purchase. To our knowledge
 934 no analog of this lemma appears in the performative-prediction, pricing, or experimental-design
 935 literatures (literature review items G1 to G3).

936 **Caveats recorded with the derivation.** (i) The constant in the $O(T^{-1/2})$ term is explicit,
 937 $2g_1 c^2 \sigma / (\gamma_{\text{PO}} v)$, and blows up as $v \rightarrow 0$: a policy that barely explores can have its tiny cost
 938 substantially rebated. This is consistent, since in that regime both sides of the product are domi-
 939 nated by the transient (D1); the theorem is stated for stationary-scale exploration, $S_T = \Theta(T)$. (ii)
 940 $g_1 = \beta |h^* - \psi|$ is $O(1)$ in general; in the weak-performativity regime it is $O(\varepsilon)$, giving the sharper
 941 factor $(1 - O(\varepsilon T^{-1/2}))$. (iii) The fixed- δ regime is different, and the certainty-equivalent anchor
 942 is the reason. At fixed prior separation the policy eventually learns s outright and can exploit it

indefinitely: the exploitation fraction does not vanish (observed directly in V3b’s first, deliberately mis-scaled run). That regime is not a counterexample. Sustained exploitation of learned structure is exactly what D0’s certainty-equivalent anchor re-anchors away: once ε is known, the CE-optimal deployment shifts, and the “gain” is no longer measured as a rebate against exploration. The theorem’s content concerns the unlearned residual, whose hardest instances live at the minimax scale $\delta \sim \sigma/\sqrt{S_T}$, and there the rebate vanishes at rate $T^{-1/2}$ (verified). We state this distinction explicitly: it separates the theorem’s content from an apparent counterexample.

Numerical verification (V3). Three checks. (V3a) A Bayes-risk simulation with a Gaussian-prior posterior-mean estimator under three designs (front-loaded, spread, adaptive-greedy) with capped budget D : the measured risk respects $\sigma^2/(D + \sigma^2/\sigma_\pi^2)$ in every arm, with the spread design near-tight. (V3b-i) The Pinsker chain: the measured posterior imbalance satisfies $|\mathbb{E}[s | \mathcal{F}_t]| \leq \delta\sqrt{S_t}/\sigma$ pathwise along simulated runs. (V3b-ii) An explicitly exploiting policy, which drifts by the posterior mean of the unknown linear term, simulated at several horizons T : its realized gain_T/C_T decays consistently with $O(T^{-1/2})$, and its achieved $\text{Var}(\hat{\varepsilon}) \times \text{cost}$ never falls below $\frac{1}{2}\gamma_{\text{PO}}\sigma^2(1 - \text{measured rebate})$.

Structure-proofness within reach (T9). The floors above are nonparametric. A separate threat is a policy that knows the response family’s functional form $\tau(h) = C_0 + C_1 e^{-ch}$ and shops for designs whose parametric Cramér–Rao product undercuts the floor. Define, for a finitely supported design μ ,

$$R(\mu) = (g^\top M(\mu)^{-1} g) \int (h - h^*)^2 d\mu, \quad M(\mu) = \int s s^\top d\mu,$$

with the sensitivity and the estimand gradient

$$s(h) = (1, e^{-ch}, -C_1 h e^{-ch})^\top, \quad g = \nabla_\theta \varepsilon(h^*) = -s'(h^*),$$

normalized so that $R(\mu) \geq 1$ says the family-knowing product is at least the exchange-rate floor $\frac{1}{2}\gamma_{\text{PO}}\sigma^2$ under the local-quadratic (A1) cost.

One-dimensional reduction. The generalized Rayleigh identity plus the bijection $u \leftrightarrow \phi_u = u^\top s$ between $\mathbb{R}^3$ and $V = \text{span}\{1, e^{-ch}, h e^{-ch}\}$ (using $C_1 \neq 0$) gives

$$g^\top M(\mu)^{-1} g = \frac{1}{\inf \{ \|\phi\|_\mu^2 : \phi \in V, \phi'(h^*) = 1 \}},$$

because $u^\top g = -\phi'_u(h^*)$ (the sign squares away). So the floor holds at μ if and only if some unit-slope family element is μ -cheaper than the linear function $d(h) = h - h^*$.

The candidate and its reach. The unique element of V with 2-jet $(0, 1, 0)$ at the anchor is

$$\phi_0(h) = \frac{2}{c} - e^{-c(h-h^*)} \left(\frac{2}{c} + (h - h^*) \right),$$

with expansion $\phi_0(x) = x - (c^2/6)x^3 + \dots$ in $x = h - h^*$. The pointwise domination $|\phi_0(x)| \leq |x|$ holds exactly on $x \in [-2/c, \infty)$ (equality only at $x = 0$ and $x = -2/c$), and fails strictly and exponentially below $-2/c$. The three elementary ingredients are $1 + t < e^t$ (which makes $\phi_0(x) - x$ strictly decreasing), $\phi'_0(x) = e^{-cx}(1 + cx) > 0$ above $-1/c$, and the single sign change of $e^{-cx}(1 + cx) + 1$ on $(-2/c, 0)$ (since $e^{-cx}(1 + cx)$ is increasing there from $-e^2$ to 1). The full proof is recorded in the proofs appendix under T9; verified in V9.1.

Consequence (T9(i)). For any design supported in $[h^* - 2/c, \infty)$: $R(\mu) \geq 1$ whenever the estimand is estimable, strict as soon as μ places mass off the two equality points $\{h^*, h^* - 2/c\}$, and $\inf_\mu R(\mu) = 1$, approached by symmetric three-point designs collapsing at the anchor (at rate δ^2 in the collapse half-width δ , V9.2; the limit is exact because in 2-jet coordinates the collapsing model is the quadratic family, where every symmetric design has $R = 1$ on the nose). The trust region A4 with $r \leq 2/c$ is therefore exactly the condition under which the floor is structure-proof, not a technicality.

981 **Sharpness of the candidate, and an exact duality.** Three facts sharpen the picture, established in
 982 the adversarial verification campaign with certified numerics (the full write-up is journal material
 983 under OPEN-5). (i) The easy direction of a duality: for a finite support X let $m(X) = \max\{\phi'(h^*) : \phi \in V, |\phi(x)| \leq |x - h^*| \forall x \in X\}$; then for every weighting w of X , $R(X, w) \geq m(X)^2$, because
 984 the maximizer ψ^* scaled to unit slope is feasible in the Rayleigh infimum with $\int (\psi^*/m)^2 dw \leq$
 985 $\int d^2 dw/m(X)^2$. Within reach ϕ_0 certifies $m(X) \geq 1$, which is exactly T9(i); numerically the duality
 986 is tight ($\inf_w R(X, w) = m(X)^2$ by linear-programming duality, certified on random supports), so
 987 pointwise domination is not a lossy proof device: the reach question is the domination question.
 988 (ii) Single-candidate sharpness: among all unit-slope elements of V , ϕ_0 is the unique one whose
 989 domination set contains a full interval $[h^* - a, h^*]$ with $a > 0$, and the maximal such a is exactly
 990 $2/c$. The affine family with 2-jet $(0, 1, \gamma)$, $\gamma \in (-1, 0)$, trades the reach boundary outward (to
 991 $u_2(\gamma) < -2/c$ in units of $1/c$) at the proved price of vacating an interval just below the anchor, and
 992 the violating designs live exactly in the vacated gap. (iii) The below-reach failure is generic, not a
 993 boundary artifact: for every $A > 2/c$ there are nonsingular designs supported in $[h^* - A, \infty) \cap (0, \infty)$
 994 with $R < 1$ (a certified three-point example in the register market attains $R = 0.8537$). Consequence
 995 for the statement of T9: by the extended-Chebyshev property of V (any nonzero element has at most
 996 two zeros counting multiplicity), every design with at least three distinct support points automatically
 997 has nonsingular information, so the estimability caveat in T9 concerns only one- and two-point
 998 designs.

1000 **The reach is exact (T9(ii)).** Below $h^* - 2/c$ the domination fails, and the failure is realized: the
 1001 frozen witness (support $\{1.8726, 1.8665, 1.3073, 0.05\}$, weights $\{0.003329, 0.955702, 0.040697,$
 1002 $0.000272\}$ normalized, frozen anchor $h^* = 1.8461$, register market $c = 1.5$) attains

$$R = 0.8516504721831888870 \dots$$

1003 (50-digit precision; V9.2), with every support point except the 0.05 probe inside the reach: one
 1004 vanishing-weight below-reach probe breaks the floor by 15 percent. Its true-cost ratio is $1.011 > 1$,
 1005 so the violation lives in the local-quadratic cost model, which is the floor's own scope. Symmetric
 1006 pair-plus-far-probe families never violate (their ratio tends to 1 from above as the probe weight
 1007 vanishes; the old scan's verdict of global structure-proofness was a weight-grid artifact, recorded as
 1008 the eleventh measurement-forced pivot). The characterization of the full below-reach violation set is
 1009 OPEN-5.

1010 **D4/D5: Design geometry: optimal shapes, dispersion, temporal shaping, Chebyshev support** 1011 **(T5, C5.1, L3, T6)**

1012 This subsection records the derivations behind the three exact design optima of T5, the dispersion
 1013 corollary C5.1, the temporal-shaping collapse L3, and the Chebyshev counting result T6; it covers
 1014 both register slots D4 and D5, the Chebyshev material being the D5 portion. All three design
 1015 optima are derived with proofs, the temporal-shaping lemma and the counting result are proved, and
 1016 everything is verified in the V4 and V5 numerical checks. It is a derivation record: each optimum is
 1017 computed in full, the caveats are stated where they arose, and the numerical anchors are those of V4
 1018 and V5. The formal statements live in the theorem register under the IDs above.

1019 **Setting.** The decision is d -dimensional and the estimand is the response Jacobian. Exploration has
 1020 stationary second moment $M := \mathbb{E}[dd^\top]$ (positive semidefinite), the response noise has variance σ^2
 1021 per coordinate, and by the d -dimensional form of the cost equivalence lemma L1 (from D0) the value
 1022 cost rate is

$$\frac{1}{2} \mathbb{E}[d^\top \Gamma_{\text{PO}} d] = \frac{1}{2} \text{tr}(\Gamma_{\text{PO}} M),$$

1023 so the value budget over horizon T is $B = (T/2) \text{tr}(\Gamma_{\text{PO}} M)$. The per-unit-time information matrix
 1024 is M/σ^2 . Absorbing T and σ^2 into units, all three problems take the static normal form of a linear
 1025 budget constraint $\text{tr}(\Gamma_{\text{PO}} M) \leq B$ on the design M .

1026 **A-optimality (T5(a)): what to do if you must know everything equally.** The problem and its
 1027 exact solution are

$$\begin{aligned} & \text{minimize} \quad \text{tr}(M^{-1}) \quad \text{subject to} \quad \text{tr}(\Gamma_{\text{PO}} M) \leq B, \quad M \succ 0, \\ & M^* = \frac{B}{\text{tr} \Gamma_{\text{PO}}^{1/2}} \Gamma_{\text{PO}}^{-1/2}, \quad \text{tr}((M^*)^{-1}) = \frac{(\text{tr} \Gamma_{\text{PO}}^{1/2})^2}{B}. \end{aligned}$$

1028 Derivation: the objective is strictly convex on the positive definite cone and the constraint is linear, so
 1029 the KKT conditions are sufficient. Stationarity reads $-M^{-2} + \lambda \Gamma_{\text{PO}} = 0$, hence $M = \lambda^{-1/2} \Gamma_{\text{PO}}^{-1/2}$;
 1030 the budget pins the multiplier through $\text{tr}(\Gamma_{\text{PO}} M) = \lambda^{-1/2} \text{tr} \Gamma_{\text{PO}}^{1/2} = B$, so $\lambda^{-1/2} = B / \text{tr} \Gamma_{\text{PO}}^{1/2}$,
 1031 and substituting back gives the stated value. A diagonal-case Cauchy–Schwarz check confirms the
 1032 shape: with curvatures g_i and design weights m_i ,

$$\left(\sum_i \frac{1}{m_i} \right) \left(\sum_i g_i m_i \right) \geq \left(\sum_i \sqrt{g_i} \right)^2,$$

1033 with equality at $m_i \propto g_i^{-1/2}$.

1034 The reading is: explore where the objective is flat, but only to the square-root degree. The optimal
 1035 exploration covariance is $\Gamma_{\text{PO}}^{-1/2}$, the inverse square root and not the inverse: flat directions are cheap
 1036 per unit information but not infinitely favoured, while curved directions are expensive but cannot be
 1037 abandoned when every coordinate of the Jacobian is needed.

1038 **The price of isotropic jitter (C5.1).** Isotropic exploration at the same budget, $M = (B / \text{tr} \Gamma_{\text{PO}}) I$,
 1039 is feasible with the budget tight and has risk $\text{tr}(M^{-1}) = d \text{tr}(\Gamma_{\text{PO}}) / B$. The ratio to the A-optimal
 1040 value is

$$F = \frac{d \text{tr}(\Gamma_{\text{PO}})}{(\text{tr} \Gamma_{\text{PO}}^{1/2})^2} \geq 1,$$

1041 with equality if and only if all curvatures are equal (Cauchy–Schwarz again). F is the curvature
 1042 dispersion of the objective: the exact factor by which naive isotropic exploration, which is optimal
 1043 in LQR-like settings (Simchowitz and Foster, 2020), overpays in the performative setting. This is
 1044 the content of C5.1, restated as the isotropy contrast P9.2, and F is computable on the calibrated
 1045 multi-bond Γ_{PO} (whose diagonal is given exactly by P9.3).

D-optimality (T5(b)): what the confidence-volume objective wants.

$$\begin{aligned} & \text{maximize} \quad \log \det M \quad \text{subject to} \quad \text{tr}(\Gamma_{\text{PO}} M) \leq B, \\ & M^* = \frac{B}{d} \Gamma_{\text{PO}}^{-1}, \quad \log \det M^* = d \log \frac{B}{d} - \log \det \Gamma_{\text{PO}}. \end{aligned}$$

1046 Derivation: $\nabla \log \det M = M^{-1}$, so stationarity reads $M^{-1} = \lambda \Gamma_{\text{PO}}$; the objective is concave and
 1047 the constraint linear, and the budget gives $\lambda^{-1} = B/d$.

1048 A caveat recorded at derivation time: the two shapes differ, A-optimality giving $\Gamma_{\text{PO}}^{-1/2}$ and D-
 1049 optimality giving Γ_{PO}^{-1} , and the two must never be conflated in the paper. The algorithm derivation
 1050 (D6) uses the D-optimal criterion internally (Frank–Wolfe on $\log \det$), while the headline risk
 1051 statement is A-optimal; both are stated, and the discrepancy factor between them is itself computable
 1052 and is small when the dispersion F is moderate.

1053 **c-optimality (T5(c)): what the corrected (PerfGD-style) update actually needs.** This is the new
 1054 result of the three. The corrected update consumes the response estimate through a single functional,
 1055 the correction direction c : in the scalar family the correction Δ needs only $\varepsilon(h^*)$, and in d dimensions
 1056 $c = \nabla_E \Delta$ is a fixed vector once the operating point is known. Estimating the scalar $c^\top \theta$ alone:

$$\begin{aligned} & \text{minimize} \quad c^\top M^{-1} c \quad \text{subject to} \quad \text{tr}(\Gamma_{\text{PO}} M) \leq B, \\ & M^* = B \frac{c c^\top}{c^\top \Gamma_{\text{PO}} c} \quad (\text{rank one: probe along } c \text{ itself}), \quad c^\top (M^*)^{-1} c = \frac{c^\top \Gamma_{\text{PO}} c}{B}. \end{aligned}$$

1057 Derivation: substitute $N = \Gamma_{\text{PO}}^{1/2} M \Gamma_{\text{PO}}^{1/2}$, so that $\text{tr } N = \text{tr}(\Gamma_{\text{PO}} M) \leq B$, and $w = \Gamma_{\text{PO}}^{1/2} c$; the
 1058 objective becomes $w^\top N^{-1} w$. By Cauchy–Schwarz,

$$(w^\top w)^2 = (w^\top N^{-1/2} N^{1/2} w)^2 \leq (w^\top N^{-1} w)(w^\top N w),$$

1059 and $w^\top N w \leq \|w\|^2 \text{tr } N \leq \|w\|^2 B$, so $w^\top N^{-1} w \geq \|w\|^2 / B$, attained at $N^* = B w w^\top / \|w\|^2$.
 1060 Back-substituting,

$$M^* = \Gamma_{\text{PO}}^{-1/2} N^* \Gamma_{\text{PO}}^{-1/2} = B \frac{(\Gamma_{\text{PO}}^{-1/2} w)(\Gamma_{\text{PO}}^{-1/2} w)^\top}{\|w\|^2}, \quad \Gamma_{\text{PO}}^{-1/2} w = c, \quad \|w\|^2 = c^\top \Gamma_{\text{PO}} c,$$

1061 giving $M^* = B c c^\top / (c^\top \Gamma_{\text{PO}} c)$ as stated.

1062 The corrected trio is worth displaying side by side:

$$\text{A-opt: } M^* \propto \Gamma_{\text{PO}}^{-1/2}, \quad \text{D-opt: } M^* \propto \Gamma_{\text{PO}}^{-1}, \quad \text{c-opt: } M^* \propto c c^\top.$$

1063 A-optimality shapes exploration as $\Gamma_{\text{PO}}^{-1/2}$, D-optimality as Γ_{PO}^{-1} , and c-optimality probes along the
 1064 correction direction itself: the curvature does not tilt the direction at all, it only prices it. The value
 1065 $c^\top \Gamma_{\text{PO}} c / B$ says that the cost of knowing your correction is the Γ_{PO} -norm of what the correction
 1066 asks for. The savings over the A-optimal value is the operational argument for correction-targeted
 1067 probing: a desk does not need the whole Jacobian, and the geometry says exactly what it may skip.

1068 Two caveats, recorded as they arose. First, an earlier draft of this derivation mis-stated the c-optimal
 1069 direction as $\Gamma_{\text{PO}}^{-1} c$; the substitution above corrects it. The error is recorded here per the project’s
 1070 falsification convention, and the corrected direction is locked by the V4 numerical check. Second,
 1071 rank-one designs are singular (hence the pseudo-inverse in the value); in practice they are regularized
 1072 by the trust region, and the paper states both the limiting and the regularized forms.

1073 **Temporal shaping is irrelevant (L3): design space collapse. Lemma L3 (temporal shaping**
 1074 **is irrelevant).** With response noise i.i.d. and independent of the deployment path, the information
 1075 about the response parameters depends on the exploration process only through its empirical second
 1076 moment $\sum_t d_t d_t^\top$. The computation is one line: the conditional log-likelihood is

$$\sum_t -\frac{(\tau_t - a - \theta^\top d_t)^2}{2\sigma^2},$$

1077 and its Hessian in θ is $-\sum_t d_t d_t^\top / \sigma^2$, regardless of the temporal order or dependence structure of
 1078 $\{d_t\}$. Hence temporally correlated exploration (AR-shaped, cyclic, burst) buys nothing beyond its
 1079 stationary design measure, and the design problem is exactly the static problems treated above. Scope
 1080 boundary: the lemma fails when the noise is serially correlated or feedback-coupled (the regime
 1081 of section 4 of D2), where shaping does matter; this is stated as the boundary of L3 and left to the
 1082 journal version.

1083 **Chebyshev counting (T6): why curvature is invisible to trajectories.** This is the D5 portion of
 1084 the record. The structural family of REFLEX theory 1.1 (`structural_response.py`) is

$$\tau(h) = C_0 + C_1 e^{-ch}, \quad \text{parameters } (C_0, C_1, c), \quad p = 3,$$

1085 with sensitivity vector

$$s(h) = (1, e^{-ch}, -C_1 h e^{-ch})^\top.$$

1086 (In this paragraph c is the decay rate of the structural family, not the correction direction of the
 1087 c-optimal problem.) The counting argument has four parts.

1088 (i) $\{1, e^{-ch}, h e^{-ch}\}$ is an extended Chebyshev system on any interval: the relevant Wronskian-type
 1089 determinants are nonzero (standard).

1090 (ii) Consequently the Fisher matrix $M(\xi) = \sum_i w_i s(h_i) s(h_i)^\top$ of any design ξ with fewer than
 1091 three distinct support points is singular, and the decay rate c is unidentified. The computation:
 1092 $\text{rank } M(\xi) \leq \#\text{support}(\xi)$, so a two-point design gives rank at most $2 < 3 = p$, hence $\det M(\xi) =$
 1093 0 and the c -direction lies in (or meets) the null space. The exact-arithmetic check in V5 finds

1094 $\det M = 0$ to machine precision for every random two-point design and $\det M > 0$ for generic
1095 three-point designs.

1096 (iii) Any noiseless RRM trajectory has effective support at most two (C1.2 of D1: geometric collapse
1097 for modulus below one, a period-two orbit at the boundary). Hence no retraining trajectory, at any
1098 modulus, identifies the curvature of its own response family.

1099 (iv) D-optimal designs for this family on a trust-region interval $[h^* - r, h^* + r]$ are supported on
1100 exactly three points including both endpoints (a de la Garza-type argument; the interior point is
1101 computed numerically in the experiments).

1102 This is the sharp form of the repository’s empirical rule that exponential response fits need
1103 a wide spread range, and the formal reason the anti-echo freeze (the identified flag in
1104 `structural_response.py`) exists.

1105 **Verified numerically (V4, V5).** Six checks anchor the results above. (1) A-optimality: random
1106 search over positive semidefinite M at fixed budget never beats $(\text{tr } \Gamma_{\text{PO}}^{1/2})^2/B$, and the closed-form
1107 M^* achieves it (run at $d = 5$ with random symmetric positive definite Γ_{PO}). (2) D-optimality:
1108 $M^* = (B/d) \Gamma_{\text{PO}}^{-1}$ beats random feasible designs in $\log \det$. (3) c-optimality: random search
1109 never beats $c^\top \Gamma_{\text{PO}} c/B$, and the rank-one construction approaches it. (4) The isotropic factor
1110 $F = d \text{tr}(\Gamma_{\text{PO}})/(\text{tr } \Gamma_{\text{PO}}^{1/2})^2$ matches the measured ratio. (5) Temporal lemma: AR-shaped and i.i.d.
1111 exploration with matched empirical second moment give identical OLS covariance. (6) Chebyshev:
1112 $\det M = 0$ for two-point designs versus $\det M > 0$ for three-point designs, and the c -direction
1113 variance is infinite or huge under two-point designs.

1114 **D6: Safe certainty equivalence (L4, R2, P7.1)**

1115 This derivation records the safety analysis of the certainty-equivalent corrected loop: the structural
1116 observation R2 (open-loop exploration cannot destabilize the linearized retraining dynamics), the
1117 perturbed-modulus lemma L4 (how C^1 estimation error in the response moves the closed-loop
1118 modulus), and the safety proposition P7.1 (the pessimism gate inherits uniform-in-time safety from
1119 an anytime-valid confidence sequence). The formal statements are recorded in the register under
1120 these IDs, and the proofs in the proofs appendix. Status of the record: R2 and L4 are derived and
1121 verified (V6); P7.1 is proved conditional on the validity of the confidence sequence; the $O(\sqrt{T})$
1122 design-regret bound in the closing paragraph is stated with a proof strategy only and is labeled a
1123 journal-version item.

1124 **Where the stability constraint actually bites (R2).** A fact discovered in the derivation and
1125 promoted to the paper: open-loop exploration cannot destabilize the linearized loop. The linearized
1126 deviation dynamics under a planned (history-independent) probe sequence (u_t) read

$$d_{t+1} = -m d_t + u_t,$$

1127 and the homogeneous part keeps its modulus $m < 1$ regardless of the input u : bounded inputs give
1128 bounded outputs. This is R2. The stability risk of probing therefore enters through exactly two
1129 channels.

1130 First, estimate feedback, the real one. The corrected update deploys

$$h_{t+1} = h_t + \eta \hat{\Phi}'(h_t), \quad \hat{\Phi}' = G + \hat{\Delta},$$

1131 where $\hat{\Delta}$ is built from the estimated response $\hat{\varepsilon}(\cdot)$. Estimation error changes the closed-loop map
1132 itself; this is where safety must be engineered, and it is the subject of L4.

1133 Second, basin exit, the nonlinear channel: large probes can leave the contraction neighbourhood.
1134 This is handled by the trust region $|d_t| \leq r$, a constraint the design problems of D4 already carry.

1135 **The perturbed-modulus lemma (L4).** Let the corrected map be

$$\Psi(h) = h + \eta [G(h) - \beta(h - \psi) \hat{\varepsilon}(h)],$$

1136 let $\hat{\rho}$ be its slope at the fixed point, and let ρ be the slope under the true response $\varepsilon(\cdot)$, namely
1137 $\rho = 1 - \eta \gamma_{\text{PO}}$, the Newton-scaled contraction of the corrected update [Nagarajan and Ashok, 2026].

1138 **Lemma L4 (perturbed modulus).** With $e(h) := \hat{\varepsilon}(h) - \varepsilon(h)$ the C^1 estimation error,

$$|\hat{\rho} - \rho| \leq \eta \beta \left(\|e\|_\infty + |h^* - \psi| \|e'\|_\infty \right) =: \eta \beta L_{\text{corr}} \|e\|_{C^1}.$$

1139 The computation is short. The slope of the corrected map is $\Psi'(h) = 1 + \eta \hat{\Phi}''(h)$, so the modulus
1140 perturbation is η times the curvature error, and

$$\begin{aligned} \hat{\Phi}''(h) - \Phi''(h) &= -\beta \frac{d}{dh} \left[(h - \psi) e(h) \right] \\ &= -\beta \left[e(h) + (h - \psi) e'(h) \right]. \end{aligned}$$

1141 Taking absolute values and the supremum over the operating interval gives the bound.

1142 For the structural family the C^1 error is controlled by the parameter error with explicit constants:
1143 with $e = \varepsilon(\cdot; \hat{\theta}) - \varepsilon(\cdot; \theta)$,

$$\|e\|_{C^1} \leq L_\theta \|\hat{\theta} - \theta\|,$$

1144 where L_θ is computable from (C_1, c, r) . Consequently a confidence region for θ translates directly
1145 into an interval for the closed-loop modulus.

1146 **The pessimism rule and the safety proposition (P7.1).** Maintain any anytime-valid confidence
1147 sequence Θ_t for the response parameters (from common-random-number machinery as in Nagarajan
1148 and Ashok [2026], or a standard self-normalized bound; its width shrinks as information accumulates,
1149 per D2).

1150 **Rule.** At each step choose the pair (step size η_t , design) so that the worst-case modulus over Θ_t
1151 respects the margin,

$$\sup_{\theta \in \Theta_t} |\hat{\rho}(\theta)| \leq 1 - c_{\text{margin}},$$

1152 that is, pessimistic on stability, while the design objective of D4 is evaluated optimistically. When
1153 Θ_t is too wide to certify the margin, the rule freezes the correction and explores under the blind
1154 loop, which is provably stable with modulus $m < 1$ by R2. This is exactly the anti-echo freeze of
1155 `structural_response.py`, now derived rather than engineered: this freeze is the pessimism gate.

1156 **Proposition P7.1 (safety).** If the confidence sequence is valid at level $1 - \delta$ uniformly over time, then
1157 with probability at least $1 - \delta$ the realized modulus satisfies $|\rho_t| \leq 1 - c_{\text{margin}}$ for all t simultaneously.
1158 The argument is one line: on the event that $\theta \in \Theta_t$ for every t , which has probability at least $1 - \delta$ by
1159 anytime-validity, the rule's supremum dominates the realized modulus at every step. The engineering
1160 reading: safety is inherited from the confidence sequence's uniform validity; there is no union bound
1161 over steps and no per-step failure accumulation.

1162 **Design regret (stated; journal deliverable).** **Claim (to be proved in full).** The certainty-equivalent
1163 re-solving scheme (the D-optimal objective from D4, the pessimism rule above, and the trust region r)
1164 attains identification regret $O(\sqrt{T})$ against the budget-matched oracle design. The elliptical-potential
1165 argument of linear-bandit design theory supplies the rate, and the freeze episodes contribute an
1166 additive $O(\log T)$ term: each freeze ends when the confidence width halves, and the widths shrink
1167 geometrically in the accumulated design energy. As a remark on scope, this item is labeled partial for
1168 the workshop version: the algorithm and the safety guarantee are proved, while the regret constant
1169 and the freeze-episode accounting are the journal version's work.

1170 **Numerical verification (V6).** Three checks, all in V6.

- 1171 • Open-loop neutrality: injected bounded probe schedules never change the measured ho-
1172 mogeneous contraction rate, read off as the slope of the deviation envelope, confirming
1173 R2.
- 1174 • The perturbed-modulus lemma: corrected loops were run with deliberately corrupted $\hat{\varepsilon}$,
1175 the C^1 error swept over a grid; in every cell the measured fixed-point slope deviates from
1176 $1 - \eta \gamma_{\text{PO}}$ by less than the bound of L4, and the bound is tight within a factor of about two
1177 at small errors.
- 1178 • The pessimism rule: under a simulated valid confidence sequence, no trajectory in 400 seeded
1179 runs violates the margin; under a deliberately invalid (too-narrow) sequence, violations
1180 appear, so the validity assumption of P7.1 is necessary, not an artifact of the proof.

1181 **D7: The anchoring crossover (T7)**

1182 This subsection records the derivation behind T7: when does anchoring the response estimate to the
 1183 structural family (“structure”) beat a free-form nonparametric estimator (“capacity”) ? The headline
 1184 of the derivation record is that the crossover is horizon-dependent, not budget-only. The formal
 1185 statement is recorded as T7 in the theorem register. A status remark carried over from the derivation
 1186 record: the derivation is complete, including the subtlety that changes the statement of the theorem
 1187 (the nonparametric bias floor is horizon-dependent), and both the exact secant-bias constant and the
 1188 two MSE formulas below are verified numerically in V7.

1189 **The question.** REFLEX’s central empirical finding is that anchoring beats a free-form estimator:
 1190 the v3/v4 negative-then-positive result. When is that the right choice *in general*? The naive answer
 1191 prices misspecification against the exchange rate of T2 alone:

$$\text{anchor} \quad \text{iff} \quad \delta_{\text{mis}}^2 < \frac{\gamma_{\text{PO}} \sigma^2}{2B}.$$

1192 Working the derivation shows the naive answer is incomplete: the nonparametric alternative’s bias is
 1193 horizon-dependent, and the honest crossover involves $(\delta_{\text{mis}}, B, T)$ jointly.

1194 **The nonparametric side: local (secant) slope estimation.** Estimate $\varepsilon = -\tau'(h^*)$ from symmetric
 1195 probes at $h^* \pm w$, using n probe pairs, under the value budget and the horizon. Writing $S = 2nw^2$
 1196 for the design energy, the two constraints read

$$\begin{aligned} \text{budget:} \quad & \frac{1}{2} \gamma_{\text{PO}} S \leq B, \\ \text{horizon:} \quad & 2n \leq T \implies w^2 = \frac{S}{2n} \geq \frac{S}{T}, \end{aligned}$$

1197 and with the budget binding, $S = 2B/\gamma_{\text{PO}}$, the horizon forces fat probes:

$$w^2 \geq \frac{2B}{\gamma_{\text{PO}} T}.$$

1198 Secant bias, with its exact leading constant. For the symmetric difference quotient,

$$\frac{\tau(h^* + w) - \tau(h^* - w)}{2w} = \tau'(h^*) + \frac{\tau'''(h^*)}{6} w^2 + O(w^4):$$

1199 the even-order terms cancel by symmetry, so the second derivative does not enter at all. Symmetric
 1200 probing is therefore already bias-optimal at its order, a point worth one sentence in the paper. For the
 1201 structural family $\tau(h) = C_0 + C_1 e^{-ch}$ one has $\tau'''(h) = -c^3 C_1 e^{-ch}$, so

$$\text{bias}(w) = -\frac{c^3 C_1 e^{-ch^*}}{6} w^2 = -\frac{c^2}{6} \varepsilon w^2,$$

1202 using $\varepsilon = \varepsilon(h^*) = -\tau'(h^*) = c C_1 e^{-ch^*}$: a curvature multiple of the slope itself, with the sign fixed
 1203 by the family (only the magnitude enters the MSE below).

1204 **MSE at the budget- and horizon-constrained optimum.** Each probe pair yields a difference quotient
 1205 with variance $\sigma^2/(2w^2)$; averaging the n pairs gives

$$\text{Var} = \frac{\sigma^2}{2nw^2} = \frac{\sigma^2}{S} = \frac{\gamma_{\text{PO}} \sigma^2}{2B}$$

1206 at the binding budget, independent of how S splits into n and w . The bias, in contrast, requires small
 1207 w , and the horizon caps how small: $w^2 \geq 2B/(\gamma_{\text{PO}} T)$. Hence, at the constrained optimum,

$$\text{MSE}_{\text{np}} = \frac{\gamma_{\text{PO}} \sigma^2}{2B} + \left(\frac{\tau'''(h^*)}{6} \right)^2 \left(\frac{2B}{\gamma_{\text{PO}} T} \right)^2.$$

1208 The discovered subtlety, and the reason T7 carries the horizon: the bias term *falls* as T grows at fixed
 1209 budget. With more (cheaper, smaller) probes the nonparametric estimator approaches the parametric
 1210 rate. Nonparametric identification is not uniformly worse; it is worse at *short horizons and fat probes*.

1211 **The parametric (anchored) side.** For a correctly specified family, $\text{MSE}_p = \kappa_p \gamma_{\text{PO}} \sigma^2 / (2B)$,
 1212 where $\kappa_p \geq 1$ is the design efficiency for extracting $\tau'(h^*)$ through the family; c-optimal probing in
 1213 the sense of T5(c), worked in D4, makes $\kappa_p \approx 1$, and the constant is computable. Misspecified by
 1214 δ_{mis} , the C^1 distance from the true response to the family as in D6, the anchored estimator picks up
 1215 an irreducible squared bias δ_{mis}^2 :

$$\text{MSE}_{\text{anchor}} = \kappa_p \frac{\gamma_{\text{PO}} \sigma^2}{2B} + \delta_{\text{mis}}^2.$$

1216 **The crossover.** Anchor iff $\text{MSE}_{\text{anchor}} < \text{MSE}_{\text{np}}$, that is,

$$\delta_{\text{mis}}^2 < \left(\frac{\tau'''(h^*)}{6} \right)^2 \left(\frac{2B}{\gamma_{\text{PO}} T} \right)^2 + (1 - \kappa_p) \frac{\gamma_{\text{PO}} \sigma^2}{2B};$$

1217 since $\kappa_p \geq 1$, the second term is a (small) penalty on anchoring that vanishes under c-optimal probing
 1218 through the family. The dominant first term gives the memorable form stated in T7:

$$\text{anchor} \quad \text{iff} \quad \delta_{\text{mis}} < \frac{|\tau'''(h^*)| B}{3 \gamma_{\text{PO}} T} \quad (\text{to leading order}).$$

1219 **Readings.** (i) Anchoring wins at short horizons, tight budgets, and strongly curved responses;
 1220 the free-form route wins asymptotically in T at fixed budget. (ii) The formula retrodicts the v3
 1221 negative result *quantitatively*: the free-form loop failed at modest T with the trust region forcing fat
 1222 effective probes, exactly the regime the formula assigns to anchoring. (iii) It also predicts when the
 1223 finding would reverse (long horizons, generous probe counts), a regime no REFLEX experiment tests
 1224 [Nagarajan and Ashok, 2026]: a falsifiable novel prediction. (iv) Delta over Lin–Zrnic (literature
 1225 review item G5): their plug-in analysis has neither the budget nor the horizon; this formula prices
 1226 both in the paper’s single currency.

1227 **Numerical verification (V7).** Three checks. (1) The secant-bias constant: the measured ratio
 1228 $(\text{secant} - \tau')/w^2$ converges to $\tau'''(h^*)/6$ for the exponential family (a deterministic check with
 1229 exact function evaluations, three values of w extrapolated). (2) The MSE_{np} formula: Monte-Carlo
 1230 secant estimation at the constrained optimum matches the two-term formula across a (B, T) grid.
 1231 (3) The crossover: simulated anchored fits (correct family plus injected misspecification) against the
 1232 secant estimator show an empirical crossover δ_{mis}^* tracking the formula within Monte-Carlo tolerance,
 1233 including its B/T scaling (halving T doubles the crossover threshold).

1234 **D8/D9: The ROI of self-knowledge and the LQ and multi-bond instantiations (T8, P9.1, P9.2,** 1235 **P9.3)**

1236 This subsection records the derivations behind Theorem T8 (the return on investment of self-
 1237 knowledge), the two consistency propositions P9.1 and P9.2, and the multi-bond curvature in-
 1238 stantiation P9.3. Status of the underlying derivation document: the ROI closed forms are derived and
 1239 verified (V8); both consistency propositions are proved (one-line proofs, verified); the multi-bond
 1240 Γ_{PO} is derived exactly in the separable case, with the coupled extension stated to first order. The
 1241 formal statements live in the theorem register and are quoted here only as needed for the algebra.

1242 **Ingredients.** Three constants, all computable from earlier derivations and the primitives of the
 1243 REFLEX market [Nagarajan and Ashok, 2026], enter the decision problem:

$$A = \frac{1}{2} \gamma_{\text{PO}} (h_{\text{SP}} - h_{\text{PO}})^2, \quad b = \frac{1}{2} \gamma_{\text{PO}} \sigma^2, \quad a = \frac{1}{2} \gamma_{\text{PO}} \kappa^2.$$

1244 Here A is the echo-chamber gap, the per-step value of running the corrected deployment h_{PO} instead
 1245 of the naive h_{SP} ; b is the exchange rate of T2, under which estimator accuracy v costs b/v in
 1246 performative value (derivation D2); and a is the residual gap per unit of estimator variance: an
 1247 error in $\hat{\varepsilon}$ of variance v displaces the corrected deployment by $\kappa^2 v$ in squared spread units, where
 1248 $\kappa = |dh_{\text{PO}}/d\varepsilon|$ is the sensitivity of the corrected optimum to the response slope, available in closed
 1249 form from the market model’s fixed-point equation [Nagarajan and Ashok, 2026].

1250 **The decision problem.** Explore once to accuracy v (paying the exchange-rate price b/v up front),
 1251 then run the corrected deployment forever, discounting the perpetual stream at rate ρ_{disc} . The
 1252 corrected stream is worth $A - av$ per step, so the net present value is

$$\text{NPV}(v) = \frac{A - av}{\rho_{\text{disc}}} - \frac{b}{v}, \quad v \in (0, \infty).$$

1253 **Optimal accuracy and break-even patience (T8).** Differentiating,

$$\text{NPV}'(v) = -\frac{a}{\rho_{\text{disc}}} + \frac{b}{v^2} = 0 \quad \implies \quad v^* = \sqrt{\frac{b \rho_{\text{disc}}}{a}} = \frac{\sigma}{\kappa} \sqrt{\rho_{\text{disc}}},$$

1254 where the second form follows by substituting a and b : the common factor $\frac{1}{2}\gamma_{\text{PO}}$ cancels, so the
 1255 optimal accuracy does not involve the curvature at all. Since $\text{NPV}''(v) = -2b/v^3 < 0$ on $(0, \infty)$,
 1256 the objective is concave there and v^* is the global maximizer. At the optimum the two v -dependent
 1257 terms are equal,

$$\frac{a v^*}{\rho_{\text{disc}}} = \sqrt{\frac{a b}{\rho_{\text{disc}}}} = \frac{b}{v^*},$$

1258 so the optimal value is

$$\text{NPV}(v^*) = \frac{A}{\rho_{\text{disc}}} - 2 \sqrt{\frac{a b}{\rho_{\text{disc}}}}.$$

1259 Exploring at all is worthwhile iff $\text{NPV}(v^*) > 0$, which rearranges to

$$A > 2 \sqrt{a b \rho_{\text{disc}}} \quad \iff \quad \rho_{\text{disc}} < \rho^* := \frac{A^2}{4 a b} = \frac{(h_{\text{SP}} - h_{\text{PO}})^4}{4 \kappa^2 \sigma^2}.$$

1260 The last equality is the substitution $A^2/(4ab) = (\frac{1}{2}\gamma_{\text{PO}})^2 (h_{\text{SP}} - h_{\text{PO}})^4 / (4 \cdot \frac{1}{2}\gamma_{\text{PO}}\kappa^2 \cdot \frac{1}{2}\gamma_{\text{PO}}\sigma^2)$,
 1261 in which γ_{PO} cancels completely.

1262 Three readings of the closed forms. (i) The break-even patience $\rho^* = (h_{\text{SP}} - h_{\text{PO}})^4 / (4\kappa^2\sigma^2)$
 1263 involves only the deployment gap, the correction's sensitivity, and the response noise: the curvature
 1264 drops out, a clean and slightly surprising closed form (it cancels between the gap's value and the
 1265 exploration price). (ii) Everything on the right-hand side is computable from the primitives of
 1266 REFLEX theory before exploring, which is the point of the result as a decision rule. (iii) v^* is the
 1267 right amount of self-knowledge: more accuracy than v^* is vanity, less is blindness.

1268 **Frontier consistency (P9.1).** For any deterministic design d_t with energy $S_T = \sum_{t \leq T} d_t^2$, the
 1269 pathwise exchange-rate product is

$$\frac{\sigma^2}{S_T} \times \frac{1}{2} \gamma_{\text{PO}} S_T = \frac{1}{2} \gamma_{\text{PO}} \sigma^2 :$$

1270 the S_T cancels, so nothing about the schedule matters. In particular the $d_t \sim c/\sqrt{t}$ schedules with
 1271 $S_T = c^2 \log T$, the Lai–Robbins cost- $O(\log T)$ regime of adaptive design, satisfy the exchange rate
 1272 of T2 with equality: their celebrated cheapness buys proportionally little information, and they sit
 1273 exactly on the frontier.

1274 **Isotropy contrast (P9.2).** In the d -dimensional problem, the ratio of the isotropic design's risk to
 1275 the A-optimal design's risk at matched budget is the curvature-dispersion factor of D4 and C5.1,

$$F = \frac{d \operatorname{tr}(\Gamma_{\text{PO}})}{(\operatorname{tr} \Gamma_{\text{PO}}^{1/2})^2} \geq 1,$$

1276 with $F = 1$ iff Γ_{PO} is a multiple of the identity. That is exactly the LQR-like isotropic case in
 1277 which naive isotropic exploration is optimal (Simchowitz–Foster 2020); generically $F > 1$ and
 1278 grows with the spread of the curvature spectrum. The performative setting is generically anisotropic
 1279 (the bond-level curvatures γ_a span decades across the calibrated universe), so naive exploration is
 1280 generically suboptimal: this is the contrast between the two regimes stated in the main text.

1281 **Separable multi-bond curvature (P9.3, exact).** In the multi-bond market with per-bond decisions
 1282 h_a and separable flows, the response Jacobian is diagonal, $E = \operatorname{diag}(\varepsilon_a)$, and the performative
 1283 objective separates across bonds. Applying the scalar curvature computation [Nagarajan and Ashok,
 1284 2026] bond by bond gives, exactly,

$$(\Gamma_{\text{PO}})_{aa} = \gamma_a + \beta \varepsilon_a (2 + c_t \psi_a - c_t h_a), \quad (\Gamma_{\text{PO}})_{ab} = 0 \quad (a \neq b).$$

1285 All quantities on the right are the calibrated per-bond constants of the REFLEX market [Nagarajan
 1286 and Ashok, 2026]. Consequently the dispersion factor F of P9.2 is directly computable on the
 1287 calibrated universe (170 CUSIPs with at least twelve months of returns; the REALDATA record);
 1288 that computed value is the number quoted in the paper's experiments.

1289 **Factor-coupled case (construction, first order).** With factor-coupled responses, E diagonal plus
 1290 low rank as in REFLEX theory 1.5, the same differentiation at the operating point gives

$$\Gamma_{\text{PO}} = \Gamma + \beta \left[2 E_{\text{sym}} + c_t (\text{diag}(\psi) E_{\text{sym}} - H E_{\text{sym}}) \right] + O(\|E_{\text{offdiag}}\|^2),$$

$$E_{\text{sym}} = \frac{1}{2} (E + E^\top), \quad H = \text{diag}(h_a),$$

1291 which is exact when E is symmetric and is evaluated at the operating point. The low-rank structure
 1292 of E is inherited by the correction, so $\Gamma_{\text{PO}}^{-1/2}$ and $\text{tr} \Gamma_{\text{PO}}^{1/2}$ remain computable in $O(dk^2)$ via the same
 1293 Woodbury and eigen-split machinery already used in REFLEX theory 1.5: the instantiation adds no
 1294 new computational burden.

1295 Caveat (honesty convention). The paper states the separable case as an exact result (P9.3) and
 1296 the coupled case only as a first-order construction carrying its explicit $O(\|E_{\text{offdiag}}\|^2)$ error term,
 1297 matching the reporting conventions of Nagarajan and Ashok [2026].

1298 **Numerical verification (V8).** Five checks. (1) $v^* = \sqrt{b \rho_{\text{disc}}/a}$ and the $\text{NPV}(v^*)$ formula match
 1299 a direct numerical scan over v on three parameter sets. (2) Root-finding $\text{NPV}(v^*)(\rho_{\text{disc}}) = 0$ in ρ_{disc}
 1300 recovers $\rho^* = (h_{\text{SP}} - h_{\text{PO}})^4 / (4\kappa^2 \sigma^2)$ to 10^{-10} , confirming the γ_{PO} cancellation. (3) The frontier
 1301 proposition: $c/\sqrt{t}$ schedules measure exactly on-frontier. (4) The dispersion factor on a synthetic
 1302 curvature spectrum spanning two decades: F matches the measured isotropic-to-optimal risk ratio
 1303 (cross-checked against V4). (5) The separable Γ_{PO} : a finite-difference Hessian of the separable
 1304 objective matches the per-bond formula entrywise.

1305 D Selected verification rows

Table 1: Selected verification rows (full table, seeds, and code in the repository).

Claim (register)	Measured	Predicted	
Exchange rate, $m=0.3$, two amplitudes (T2)	0.01062, 0.01070	0.01056	pass
Exchange rate, $m=0.6$, two amplitudes (T2)	0.00549, 0.00569	0.00535	pass
Saturating environment (outside A1)	+16% over rate	out of scope	drift
Feedback floor, $m=0.3/0.6/0.85$ (T1 corollary; E1)	within 0.2–6.4%	$(\sigma/\gamma)^2/(1-m^2)$	pass
Design arm $\text{sd}(\hat{c})$; jitter/design ratio (T6)	0.234; $7.8\times$	Fisher 0.248; ≥ 3	pass
SafeD reaches h_{PO} ; trust region (L4)	1.258; 0 violations	1.296; 0	pass
Isotropic/A-optimal risk ratio, $d=8$ (T5)	1.529	$F = 1.506$	pass
Nonparametric MSE at (B, T) optimum (T7)	0.0132	0.0123	pass
Pricing domain, 4 cells (T2, exact: LQ); agent	0.384, 0.192; 12.65	0.384, 0.192; 12.5	pass
Multi-bond shaped/iso steady-state risk (T5)	1.20 (flat control 1.00)	$> 1.15 (\approx 1)$	pass
Within-reach min CR ratio; witness beyond (T9)	1.0005; 0.8517	≥ 1 ; < 1	pass

1306 E Additional figures

1307 F Reproducibility

1308 All results are deterministic from seeds, numpy-only, CPU-only; reported final errors are 300-step
 1309 tail averages over the deployed path, stable within 0.01 under horizon perturbation (worst measured
 1310 spread 0.0062; the repository’s stability report audits every cell at five horizons). The repository
 1311 contains: the derivation documents and theorem register (26 results; 38 numerical derivation checks),
 1312 the pipeline (environments, estimators, designs, agents, baselines; 36 measured-vs-predicted rows;
 1313 9 unit tests), the real-data leg with its 10/10 port validation, and continuous integration running the
 1314 verification suites on every push. The register of eleven measurement-forced pivots, each recorded in
 1315 the code where it happened, with the failed prediction and the fix, is in the repository README.

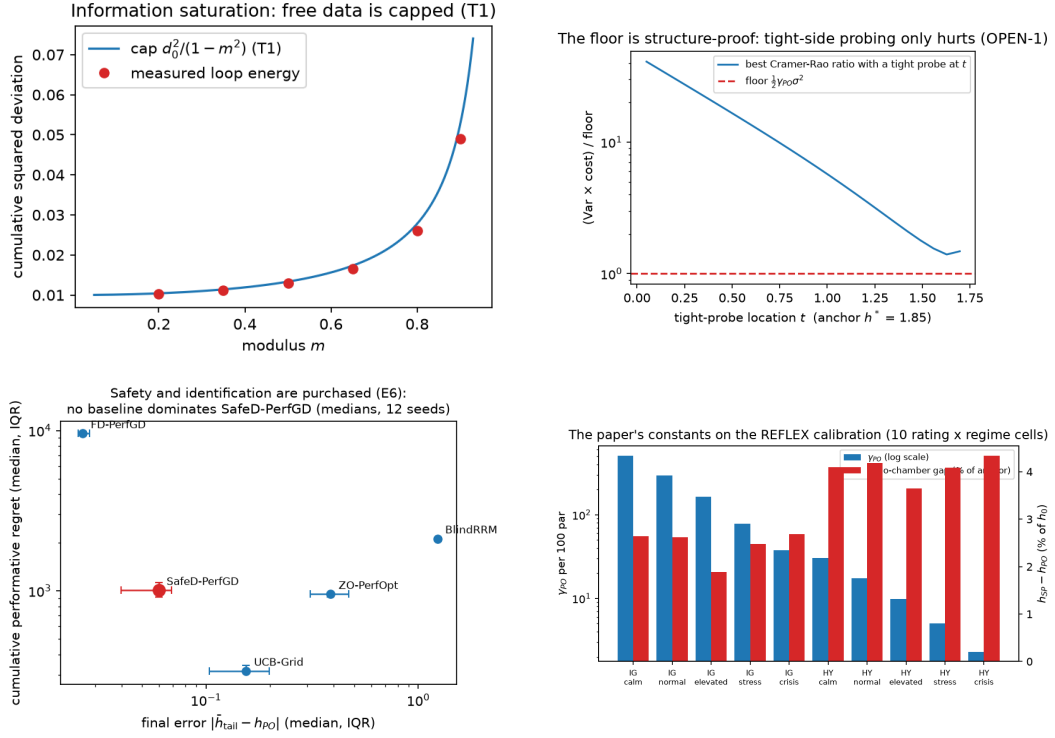


Figure 2: **Top left:** information saturation: measured loop energy against the cap $d_0^2/(1-m^2)$ (T1). **Top right:** best Cramér–Rao ratio with one tight probe at t over the coarse weight grid ($w \geq 0.05$): at appreciable weight, tight probes only hurt. The below-reach violation of Theorem 3(ii) lives outside this family (a vanishing-weight probe, 3×10^{-4} , inside an asymmetric cluster), which is exactly why the coarse scan missed it. **Bottom left:** baseline Pareto frontier: no baseline dominates SafeD-PerfGD on median (final error, regret), 12 seeds. **Bottom right:** the paper's constants on the REFLEX calibration, 10 rating $\times$ regime cells.

1316 NeurIPS Paper Checklist

1317 1. Claims

1318 Question: Do the main claims made in the abstract and introduction accurately reflect the
1319 paper’s contributions and scope?

1320 Answer: [Yes]

1321 Justification: The abstract’s claims (an exact conservation law, minimax lower bounds, a
1322 certified adaptive procedure, 36 preregistered checks with 0 failures, 10/10 calibration cells
1323 within 0.8%) map one-to-one to Theorems 1–3, Section 3, and the measurements reported
1324 in Section 4.

1325 Guidelines:

- 1326 • The answer [N/A] means that the abstract and introduction do not include the claims
1327 made in the paper.
- 1328 • The abstract and/or introduction should clearly state the claims made, including the
1329 contributions made in the paper and important assumptions and limitations. A [No] or
1330 [N/A] answer to this question will not be perceived well by the reviewers.
- 1331 • The claims made should match theoretical and experimental results, and reflect how
1332 much the results can be expected to generalize to other settings.
- 1333 • It is fine to include aspirational goals as motivation as long as it is clear that these goals
1334 are not attained by the paper.

1335 2. Limitations

1336 Question: Does the paper discuss the limitations of the work performed by the authors?

1337 Answer: [Yes]

1338 Justification: The Conclusion (Section 5) contains a dedicated limitations paragraph: the
1339 identity’s local scope (A1) with the measured +16% out-of-scope drift, the load-bearing
1340 trust region, the open sharp constant (OPEN-1), the matched-energy jitter comparison, and
1341 the real-data provenance caveat.

1342 Guidelines:

- 1343 • The answer [N/A] means that the paper has no limitation while the answer [No] means
1344 that the paper has limitations, but those are not discussed in the paper.
- 1345 • The authors are encouraged to create a separate “Limitations” section in their paper.
- 1346 • The paper should point out any strong assumptions and how robust the results are to
1347 violations of these assumptions (e.g., independence assumptions, noiseless settings,
1348 model well-specification, asymptotic approximations only holding locally). The authors
1349 should reflect on how these assumptions might be violated in practice and what the
1350 implications would be.
- 1351 • The authors should reflect on the scope of the claims made, e.g., if the approach was
1352 only tested on a few datasets or with a few runs. In general, empirical results often
1353 depend on implicit assumptions, which should be articulated.
- 1354 • The authors should reflect on the factors that influence the performance of the approach.
1355 For example, a facial recognition algorithm may perform poorly when image resolution
1356 is low or images are taken in low lighting. Or a speech-to-text system might not be
1357 used reliably to provide closed captions for online lectures because it fails to handle
1358 technical jargon.
- 1359 • The authors should discuss the computational efficiency of the proposed algorithms
1360 and how they scale with dataset size.
- 1361 • If applicable, the authors should discuss possible limitations of their approach to
1362 address problems of privacy and fairness.
- 1363 • While the authors might fear that complete honesty about limitations might be used by
1364 reviewers as grounds for rejection, a worse outcome might be that reviewers discover
1365 limitations that aren’t acknowledged in the paper. The authors should use their best
1366 judgment and recognize that individual actions in favor of transparency play an impor-
1367 tant role in developing norms that preserve the integrity of the community. Reviewers
1368 will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [\[Yes\]](#)

Justification: Assumptions (A1–A6) are named in each theorem statement and defined with the register; Appendix A states all register results in full, Appendix B gives complete proofs, and the one numerically supported constant is explicitly labelled open (OPEN-1) rather than claimed.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [\[Yes\]](#)

Justification: Every numeric claim is a preregistered measured-versus-predicted row in a deterministic, seeded pipeline; Section 4 and Appendices D and F describe the harness, and the supplementary repository reproduces all 36 rows with two commands.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not include experiments.
- If the paper includes experiments, a [\[No\]](#) answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

1423 (d) We recognize that reproducibility may be tricky in some cases, in which case
1424 authors are welcome to describe the particular way they provide for reproducibility.
1425 In the case of closed-source models, it may be that access to the model is limited in
1426 some way (e.g., to registered users), but it should be possible for other researchers
1427 to have some path to reproducing or verifying the results.

1428 5. Open access to data and code

1429 Question: Does the paper provide open access to the data and code, with sufficient instruc-
1430 tions to faithfully reproduce the main experimental results, as described in supplemental
1431 material?

1432 Answer: [Yes]

1433 Justification: Code, derivation documents, and all verification artifacts (results tables,
1434 adversarial-search record, real-data log, stability audit, unit tests) ship as the supplementary
1435 repository with run instructions.

1436 Guidelines:

- 1437 • The answer [N/A] means that paper does not include experiments requiring code.
- 1438 • Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- 1439 • While we encourage the release of code and data, we understand that this might not
1440 be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not
1441 including code, unless this is central to the contribution (e.g., for a new open-source
1442 benchmark).
- 1443 • The instructions should contain the exact command and environment needed to run to
1444 reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- 1445 • The authors should provide instructions on data access and preparation, including how
1446 to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- 1447 • The authors should provide scripts to reproduce all experimental results for the new
1448 proposed method and baselines. If only a subset of experiments are reproducible, they
1449 should state which ones are omitted from the script and why.
- 1450 • At submission time, to preserve anonymity, the authors should release anonymized
1451 versions (if applicable).
- 1452 • Providing as much information as possible in supplemental material (appended to the
1453 paper) is recommended, but including URLs to data and code is permitted.

1456 6. Experimental setting/details

1457 Question: Does the paper specify all the training and test details (e.g., data splits, hyperpa-
1458 rameters, how they were chosen, type of optimizer) necessary to understand the results?

1459 Answer: [Yes]

1460 Justification: Section 4 states environments, seed counts, and metrics; per-experiment hori-
1461 zons, constants, and tolerances are documented in the supplementary repository alongside
1462 the code that sets them.

1463 Guidelines:

- 1464 • The answer [N/A] means that the paper does not include experiments.
- 1465 • The experimental setting should be presented in the core of the paper to a level of detail
1466 that is necessary to appreciate the results and make sense of them.
- 1467 • The full details can be provided either with the code, in appendix, or as supplemental
1468 material.

1469 7. Experiment statistical significance

1470 Question: Does the paper report error bars suitably and correctly defined or other appropriate
1471 information about the statistical significance of the experiments?

1472 Answer: [Yes]

1473 Justification: The baseline comparison reports 12-seed medians with interquartile ranges
1474 (variability across seeds); deterministic identity checks carry preregistered relative tolerances
1475 and a horizon-stability audit (worst spread 0.0062 against tolerance 0.01).

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: All experiments are numpy-only, CPU-only, and deterministic from seeds (Appendix F); no GPU or cluster is used, and suite runtimes are documented in the repository README.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The work is theoretical and simulation-based, uses only cited public calibration data, and conforms to the NeurIPS Code of Ethics.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

1528 Question: Does the paper discuss both potential positive societal impacts and negative
1529 societal impacts of the work performed?

1530 Answer: [N/A]

1531 Justification: This is foundational theory on estimation–regret tradeoffs in performative
1532 feedback loops; it is not tied to a particular deployment, and no direct path to a societal
1533 application beyond the analyzed market-making setting is introduced.

1534 Guidelines:

- 1535 • The answer [N/A] means that there is no societal impact of the work performed.
- 1536 • If the authors answer [N/A] or [No], they should explain why their work has no societal
1537 impact or why the paper does not address societal impact.
- 1538 • Examples of negative societal impacts include potential malicious or unintended uses
1539 (e.g., disinformation, generating fake profiles, surveillance), fairness considerations
1540 (e.g., deployment of technologies that could make decisions that unfairly impact specific
1541 groups), privacy considerations, and security considerations.
- 1542 • The conference expects that many papers will be foundational research and not tied
1543 to particular applications, let alone deployments. However, if there is a direct path to
1544 any negative applications, the authors should point it out. For example, it is legitimate
1545 to point out that an improvement in the quality of generative models could be used to
1546 generate Deepfakes for disinformation. On the other hand, it is not needed to point out
1547 that a generic algorithm for optimizing neural networks could enable people to train
1548 models that generate Deepfakes faster.
- 1549 • The authors should consider possible harms that could arise when the technology is
1550 being used as intended and functioning correctly, harms that could arise when the
1551 technology is being used as intended but gives incorrect results, and harms following
1552 from (intentional or unintentional) misuse of the technology.
- 1553 • If there are negative societal impacts, the authors could also discuss possible mitigation
1554 strategies (e.g., gated release of models, providing defenses in addition to attacks,
1555 mechanisms for monitoring misuse, mechanisms to monitor how a system learns from
1556 feedback over time, improving the efficiency and accessibility of ML).

1557 11. Safeguards

1558 Question: Does the paper describe safeguards that have been put in place for responsible
1559 release of data or models that have a high risk for misuse (e.g., pre-trained language models,
1560 image generators, or scraped datasets)?

1561 Answer: [N/A]

1562 Justification: No models or datasets with misuse risk are released; the artifact is a small
1563 numerical verification pipeline over a cited public calibration.

1564 Guidelines:

- 1565 • The answer [N/A] means that the paper poses no such risks.
- 1566 • Released models that have a high risk for misuse or dual-use should be released with
1567 necessary safeguards to allow for controlled use of the model, for example by requiring
1568 that users adhere to usage guidelines or restrictions to access the model or implementing
1569 safety filters.
- 1570 • Datasets that have been scraped from the Internet could pose safety risks. The authors
1571 should describe how they avoided releasing unsafe images.
- 1572 • We recognize that providing effective safeguards is challenging, and many papers do
1573 not require this, but we encourage authors to take this into account and make a best
1574 faith effort.

1575 12. Licenses for existing assets

1576 Question: Are the creators or original owners of assets (e.g., code, data, models), used in
1577 the paper, properly credited and are the license and terms of use explicitly mentioned and
1578 properly respected?

1579 Answer: [Yes]

1580 Justification: The one external asset is the publicly available REFLEX corporate-bond
1581 calibration, cited where used (Sections 2 and 4); standard open-source scientific Python
1582 libraries are used as distributed.

1583 Guidelines:

- 1584 • The answer [N/A] means that the paper does not use existing assets.
- 1585 • The authors should cite the original paper that produced the code package or dataset.
- 1586 • The authors should state which version of the asset is used and, if possible, include a
1587 URL.
- 1588 • The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- 1589 • For scraped data from a particular source (e.g., website), the copyright and terms of
1590 service of that source should be provided.
- 1591 • If assets are released, the license, copyright information, and terms of use in the
1592 package should be provided. For popular datasets, `paperswithcode.com/datasets`
1593 has curated licenses for some datasets. Their licensing guide can help determine the
1594 license of a dataset.
- 1595 • For existing datasets that are re-packaged, both the original license and the license of
1596 the derived asset (if it has changed) should be provided.
- 1597 • If this information is not available online, the authors are encouraged to reach out to
1598 the asset's creators.

1599 13. New assets

1600 Question: Are new assets introduced in the paper well documented and is the documentation
1601 provided alongside the assets?

1602 Answer: [Yes]

1603 Justification: The released pipeline and derivation register are documented (READMEs,
1604 a preregistered results table, unit tests, and continuous integration) in the supplementary
1605 material.

1606 Guidelines:

- 1607 • The answer [N/A] means that the paper does not release new assets.
- 1608 • Researchers should communicate the details of the dataset/code/model as part of their
1609 submissions via structured templates. This includes details about training, license,
1610 limitations, etc.
- 1611 • The paper should discuss whether and how consent was obtained from people whose
1612 asset is used.
- 1613 • At submission time, remember to anonymize your assets (if applicable). You can either
1614 create an anonymized URL or include an anonymized zip file.

1615 14. Crowdsourcing and research with human subjects

1616 Question: For crowdsourcing experiments and research with human subjects, does the paper
1617 include the full text of instructions given to participants and screenshots, if applicable, as
1618 well as details about compensation (if any)?

1619 Answer: [N/A]

1620 Justification: The paper involves no crowdsourcing and no research with human subjects.

1621 Guidelines:

- 1622 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
1623 with human subjects.
- 1624 • Including this information in the supplemental material is fine, but if the main contribu-
1625 tion of the paper involves human subjects, then as much detail as possible should be
1626 included in the main paper.
- 1627 • According to the NeurIPS Code of Ethics, workers involved in data collection, curation,
1628 or other labor should be paid at least the minimum wage in the country of the data
1629 collector.

1630 15. Institutional review board (IRB) approvals or equivalent for research with human 1631 subjects

1632 Question: Does the paper describe potential risks incurred by study participants, whether
1633 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
1634 approvals (or an equivalent approval/review based on the requirements of your country or
1635 institution) were obtained?

1636 Answer: [N/A]

1637 Justification: The paper involves no research with human subjects, so no IRB review was
1638 required.

1639 Guidelines:

- 1640 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
1641 with human subjects.
- 1642 • Depending on the country in which research is conducted, IRB approval (or equivalent)
1643 may be required for any human subjects research. If you obtained IRB approval, you
1644 should clearly state this in the paper.
- 1645 • We recognize that the procedures for this may vary significantly between institutions
1646 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
1647 guidelines for their institution.
- 1648 • For initial submissions, do not include any information that would break anonymity (if
1649 applicable), such as the institution conducting the review.

1650 16. Declaration of LLM usage

1651 Question: Does the paper describe the usage of LLMs if it is an important, original, or
1652 non-standard component of the core methods in this research? Note that if the LLM is used
1653 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
1654 scientific rigor, or originality of the research, declaration is not required.

1655 Answer: [N/A]

1656 Justification: LLMs are not a component of the core methods; usage was limited to writing
1657 and editing assistance, which per the policy does not require declaration.

1658 Guidelines:

- 1659 • The answer [N/A] means that the core method development in this research does not
1660 involve LLMs as any important, original, or non-standard components.
- 1661 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
1662 be described.